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Prediction Markets as Forecasting Devices in Macro-Finance

Evidence from Kalshi on Inflation, Interest Rates, and Equity Markets

Master's Thesis

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Abstract

This thesis examines whether prices on the regulated prediction market Kalshi provide useful forecasts and how their performance changes as contract resolution approaches. It analyses the complete trade histories of contracts on monthly CPI inflation, the federal funds target rate and the daily close of the S&P 500. Contract ladders are converted into implied means and probability distributions and evaluated out of sample against prespecified time-series baselines, established target-specific external benchmarks and machine-learning models fitted to hourly market-price paths.

The full-ladder Kalshi implied mean has lower RMSE than the prespecified simple baseline for all three targets. Evidence against the external benchmarks is weaker. On 15 CPI releases, Kalshi's RMSE is 14 percent below the Cleveland Fed nowcast, but the difference is statistically inconclusive. The comparison with fed funds futures is also inconclusive on 12 meetings. For the S&P 500, E-mini futures outperform the prespecified full-ladder reading on their common 158-day sample; the difference is inconclusive only for the top-3 sensitivity reading.

No tested model improves on the full-ladder market forecast for either macroeconomic target. For the index, 12 of 18 models improve on the full ladder, but none improves on the top-3 reading. Across matched information sets, market-input models have lower RMSE than history-only models in 17 of 18 comparisons, although Holm-adjusted support is concentrated in the six index comparisons.

Calibration differs from point accuracy. The CPI ladder is too narrow in the upper tail, whereas the index ladder is too wide. Policy-rate uniformity and tail tests do not reject, but the concentration of probability on one outcome limits their informativeness. Relative-accuracy point estimates favour the two scheduled outcomes over the continuously traded index, but the short common samples do not establish a cross-target ordering.

The results support using the market's implied mean as a complementary forecast rather than as a universal replacement for established benchmarks. Implied probabilities require target-specific calibration, and the appropriate market reading depends on the contract's trading structure.

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List of Abbreviations

AR	Autoregressive
BLS	U.S. Bureau of Labor Statistics
CPI	Consumer Price Index
DM	Diebold–Mariano
EFFR	Effective Federal Funds Rate
FFR	Federal Funds Rate
FOMC	Federal Open Market Committee
FRED	Federal Reserve Economic Data
GBM	Gradient Boosting Machine
LSEG	London Stock Exchange Group
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MLP	Multilayer Perceptron
PIT	Probability Integral Transform
RMSE	Root Mean Squared Error
S&P 500	Standard & Poor's 500 Index

Chapter 1

Introduction and Research Setting

1.1 Motivation

In November 2020 the Commodity Futures Trading Commission designated Kalshi a contract market. Now, a federally regulated exchange in the United States could list binary contracts on economic events for the first time. Within a few years the platform was quoting contracts on the monthly inflation release, on the outcome of each Federal Open Market Committee meeting, and on the closing level of the S&P 500 – a continuously updated market price for quantities that survey panels, central-bank staff and commercial forecasters have spent decades trying to anticipate.

A contract ladder is a complete set of related Kalshi contracts covering the possible values of one underlying outcome. This set yields a central estimate. Moreover, it traces a probability distribution over the outcome, revised whenever someone trades.

Prediction markets are routinely described as efficient aggregators of dispersed information, but Snowberg et al. (2013) insist that any advantage over competing methods has to be demonstrated with data rather than inferred from theory, and the recent Kalshi literature has not yet closed that gap. Diercks et al. (2026) evaluate the macroeconomic contracts against statistical models and professional survey panels. What their design leaves untested is whether a flexible model fitted to the market's own price history, and restricted to the same information the market observes, can improve on the price it was trained on. Such an improvement would show that the terminal price does not exhaust the information in its own history.

Forecasting difficulty should also in theory differ across targets. A prediction market can add value when traders aggregate information that is not already reflected in established forecasts or prices. For an equity index, the efficient-markets hypothesis of Fama (1970) implies limited scope for an additional public-information forecast, because stock prices should incorporate that information rapidly. Consistent with this constraint, Welch and Goyal (2008) find that almost no proposed predictor of the equity premium survives out of sample. Kalshi traders may therefore struggle to forecast the S&P 500 more accurately than the stock market itself. Grossman and Stiglitz (1980), however, argue that some predictability must remain; otherwise, investors would have no incentive to collect information.

A scheduled policy-rate decision offers a different setting. Its timing and decision maker are known, and traders can update a clearly defined forecast as speeches and macroeconomic releases arrive. Monthly CPI inflation lies between the policy decision and the equity index: it is uncertain but driven by observable economic conditions and

is already covered by an institutional nowcast published every business day.

The thesis uses these contrasts to examine three targets within a common conceptual framework. The implementation applies target-specific benchmarks, samples, timing rules and model-training choices. Kalshi is compared with prespecified time-series baselines, established external benchmarks for each target, and machine-learning models trained on its own price paths. The analysis asks whether the market mean is accurate, whether the implied distribution is calibrated, whether the price path adds information beyond target history, and how forecast performance changes over a contract's life.

1.2 Prediction Markets and the Kalshi Platform

A prediction market is an exchange on which participants trade contracts whose payoff depends on a future event, so that the price can be read as the market's collective assessment of the probability that the event occurs. The case for reading it that way runs back to Hayek (1945), for whom a market price summarises information dispersed among many individuals that no participant holds in full.

For most of their modern history such markets operated in the United States on a small scale and under legal uncertainty: the Iowa Electronic Markets, the best-known early example, ran under a no-action letter that permitted it to operate as a research project with strict stake limits. Arrow et al. (2008) argued that this environment held back a tool of genuine value for policy and business and called for safe-harbour arrangements.

Kalshi, founded in 2018 and designated a contract market in 2020, is that argument realised at commercial scale: a licensed exchange with a per-contract fee, no legally binding stake cap, and maximum exposure per market reaching several million dollars.

Two design features separate it from the markets on which much of the academic literature was built. Under the July 2026 fee schedule, the general trading fee is proportional to $p(1-p)$, where p is the contract price in dollars, with market-specific multipliers and rounding rules (KalshiEX LLC, 2026). The fee is largest when the price is close to \$0.50.

Trading also runs through a quote-driven order book rather than the centralised double auction of the Iowa markets, which distinguishes the party posting an offer from the party accepting it. Bürgi et al. (2025) exploit that distinction on transaction-level data and report prices that are informative and grow more accurate towards resolution, but that display a favourite–longshot bias, in which cheap contracts win less often than their price implies. That means traders may overestimate unlikely outcomes.

Each contract is a binary option paying one dollar if the stated event occurs and nothing otherwise, trading between one and ninety-nine cents. For a continuous number such as inflation or another macroeconomic release, Kalshi can provide several contracts: the platform lists a family covering successive ranges of the same outcome, which can be read as points on a cumulative distribution function and converted into a distribution and its moments. Together, these contracts form the contract ladder and make the market usable for forecasting a continuous quantity.

The conversion, set out in Chapter 3, follows Diercks et al. (2026) and is not novel: Gürkaynak and Wolfers (2006) recovered a distribution from digital options on a single data release two decades earlier, in a market for economic derivatives that has since closed.

1.3 The Three Target Variables

Each target carries its own external benchmark, and the three differ in kind: an institutional model forecast, a futures-implied policy expectation, and a quote from a related asset market. Since those differences condition what a comparison with the market can establish, they are set out alongside the variables themselves.

The consumer price index measures the change in prices paid by urban consumers for a representative basket; the headline month-over-month change is the target used here. Its value is unknown until the Bureau of Labor Statistics publishes it and it moves substantially from month to month, yet it is driven by economic forces that a model trained on the historical series might in principle learn to anticipate. The external benchmark is the Federal Reserve Bank of Cleveland's inflation nowcast, a mixed-frequency model that combines daily energy prices with weekly and monthly indicators and is published every business day (Knotek & Zaman, 2017); the analysis uses the latest value released before each forecast cutoff.

The federal funds target rate is set by the Federal Open Market Committee at its scheduled meetings. Its value is a policy choice rather than a measurement, it moves in discrete twenty-five-basis-point steps, and it frequently does not move at all, so its own history is a poor guide to the next decision while the deliberation preceding that decision is closely watched. The external benchmark is the 30-day federal funds futures contract, the conventional market-based measure of policy expectations (Gürkaynak et al., 2007). Section 2.5 explains how the futures-implied effective rate is converted to the target-range upper bound; Chapter 6 discusses the unadjusted futures risk premium.

The S&P 500 differs from both in being itself an asset price, set in a deep and liquid market by forward-looking participants, which makes it the hardest of the three to forecast and gives it the role of a control at that end of the range.

Its external benchmark takes more construction than a quoted price suggests, because an E-mini futures quote is not a forecast of the cash index: the two differ by financing, dividends and time to expiry. The implementation therefore anchors the benchmark on the previous official cash close and adds the eligible pre-cutoff futures move, as detailed in Section 2.5. Because the licensed intraday record begins part-way through the test period, comparisons involving it use a shorter common sample.

1.4 Research Questions and Hypotheses

Four questions organise the analysis. How accurate are the market's point forecasts and its implied probability distributions? Do prices contain useful information that is not already available elsewhere? Does the market's value depend on the kind of variable being forecast? And how does its accuracy develop over the life of a contract? Each is answered by a group of hypotheses that keeps the H1–H5 labels used in the chapters that follow.

Two conventions define the main Kalshi forecast. The primary reading is the full-ladder implied mean at the final cutoff before the event. A second reading, restricted to the three most heavily traded strikes, is reported as a sensitivity analysis. It matters most for the S&P 500, where trading concentrates in a few near-the-money brackets while outer brackets can retain stale prices.

The three targets share one analytical framework but differ in implementation: hypotheses, loss functions and test procedures are common, while validation blocks, refit

cadences, horizon grids and event counts are adapted to each target's calendar and sample size, and every adaptation is documented where it arises.

Table 1.1 maps the questions onto the hypotheses and records the kind of evidence each provides. H1, H2 and H3 are the primary analyses, with prespecified comparisons and multiplicity corrections applied within each family. H4 and H5a are descriptive, H5b is a robustness check on the modelling design rather than a claim about the market, and H5c is exploratory and carries no formal test.

Table 1.1: Research Questions and Hypotheses

Question	Hypothesis	Comparison	Evidence
Q1	H1a	Market against the time-series baseline	Primary
	H1b	Market against the external benchmark	Primary
	H1c	Market against the trained machine-learning models	Primary
	H2	Implied distribution against the realised outcomes	Primary
Q2	H3a	Models with market input against the same models with history input	Primary
	H3b	Hybrid models against models with market input only	Primary
Q3	H4	Relative skill across the three target variables	Descriptive
Q4	H5a	Market error and revisions across forecast horizons	Descriptive
	H5b	Weighted training against unweighted training	Robustness
	H5c	Steepness of the horizon profile across variables	Exploratory

Notes: Research questions, the hypotheses that answer them, the comparison each hypothesis makes, and the kind of evidence it provides.

1.4.1 Point Accuracy and Calibration (Q1)

Three alternatives to the market price are available to a forecaster, and they are not equally demanding. The first is a simple model of the variable's own past, which anyone can build at no cost; beating it establishes that the price is worth reading at all.

> **H1a.** At the last cutoff before the event, the Kalshi full-ladder implied mean has lower out-of-sample squared forecast loss than the prespecified baseline based on the target's past values.

The second is the external benchmark described in Section 1.3, produced by an institution or by a related financial market with resources the event market does not have.

> **H1b.** On the exact common evaluation sample, the Kalshi full-ladder implied mean and the target-specific external benchmark have equal out-of-sample squared forecast loss.

H1b is tested with a Diebold–Mariano comparison of equal predictive accuracy. No non-inferiority margin is specified in advance or estimated, so the test can identify which forecast is the more accurate but cannot establish that two forecasts are equivalent. A rejection is reported as a difference in accuracy; a failure to reject is reported as an unresolved comparison rather than a demonstrated draw, since on a short common sample that is what low power produces whether or not a difference exists.

The third alternative is a flexible model trained on the same information the market itself observes, which is the comparison the earlier literature leaves open.

> **H1c.** No trained machine-learning model, using the same information as the market, attains a lower out-of-sample squared forecast loss than the Kalshi full-ladder implied mean.

H1c covers a family of comparisons rather than one test. Six model configurations are estimated under each of three training designs for every target, giving eighteen comparisons against the price on the same events, and the finding is reported as a count of the comparisons a model wins, with the multiplicity correction applied within the family. It is a claim about this set of models under these designs, not a general statement that the price cannot be improved. Because the external benchmarks are unavailable for some events, competitors are evaluated on different samples, so corrections are applied within each family separately rather than across all of them at once.

H1 only examines the implied mean, however an accurate mean does not guarantee an accurate probability distribution around it. A contract price is a risk-neutral probability and coincides with an average belief only under conditions that Manski (2006) questions and Wolfers and Zitzewitz (2006) defend; Chapter 2 sets out those conditions and the scoring apparatus, including the requirement of Gneiting and Raftery (2007) that a forecast distribution be as sharp as possible subject to being calibrated.

> **H2.** For each target variable, the terminal Kalshi implied distribution is calibrated: its randomised probability integral transform is uniform, its central prediction intervals attain their nominal coverage rates, and neither tail carries systematically too much or too little probability.

The direction of any failure is as informative as the failure itself, as it helps explain what kind of bias the market has. Uniformity and coverage are tested; serial independence of the transform values is not, a gap Chapter 6 takes up.

1.4.2 Incremental Information (Q2)

Equal accuracy does not imply equal information, since two forecasts can be equally accurate while drawing on different sources. The first two tests hold the model fixed and vary only what it is allowed to see.

> **H3a.** Machine-learning models trained on the Kalshi price paths have lower out-of-sample error than the same model types trained only on the target variable's history.

> **H3b.** Once a model already uses the Kalshi price paths, adding the target variable's history does not improve out-of-sample accuracy.

Together they rank the three information sets, from market-only through hybrid to history-only, and locate where the information sits. H3a compares market-only with history-only models.

H3b compares hybrid with market-only. When two forecasts disagree, the disagreement is either noise or signal; if the market knows something the history does not, the direction in which it departs from the history should predict the direction of the history's error.

1.4.3 Differences between Targets (Q3)

> **H4.** Relative to each target's conventional benchmark, the Kalshi forecast shows greater skill for the scheduled macroeconomic and policy outcomes than for the continuously traded equity index.

Because the three variables are measured in different units and differ in how hard they are to forecast, the comparison is expressed in relative rather than absolute accuracy.

Three features of the implementation make H4 descriptive rather than confirmatory. It runs on the calendar window common to all three targets, which leaves fewer than a dozen events for each; it produces a separate interval estimate for each target against that target's own benchmark, rather than a formal contrast between targets; and for the S&P 500 the market is read from the actively traded brackets rather than the full ladder, a sensitivity reading adopted after the full-ladder results had been seen. Any ordering across targets is therefore presented as a pattern in the point estimates rather than a tested result.

1.4.4 The Life of the Contract (Q4)

A contract produces a new forecast every hour, which makes the path from listing to resolution observable. Berg et al. (2008) find that market prices beat polls even at long horizons, Bürgi et al. (2025) that Kalshi prices grow more accurate towards close, and Diercks et al. (2026) document the same for scheduled macroeconomic contracts, where information arrives on a known calendar.

> **H5a.** The Kalshi forecast error and the size of forecast revisions are both smaller close to the resolution date than far from it.

H5a concerns the overall profile across the horizon grid. It does not require an improvement between every adjacent pair of cutoffs. The cutoffs are unevenly spaced, and for the S&P 500 the number of events behind a bucket reduces as the horizon lengthens, which makes strict monotonicity an unreasonable standard.

If windows recorded close to an event hold more relevant information than distant ones, a model that discounts distant windows should forecast better than one that treats them alike.

> **H5b.** Training that discounts windows recorded far from the event gives a lower terminal forecast error than training that weights all windows equally.

H5b concerns the modelling design rather than the market, which makes it a useful check: the weighting would not help out of sample if the horizon structure were just a

part of the evaluation grid. It is evaluated separately for each model configuration, the result is reported as the number of matched comparisons.

Whether the profile has the same shape for all three variables depends on how much decision-relevant information arrives during a contract's life. For example, FOMC error might fall sharply as the meeting approaches, or CPI error might fall gradually as new economic indicators appear. The horizon windows differ across targets by more than an order of magnitude, so the comparison is posed as an open question and carries no formal test.

> **H5c (exploratory).** For which target variable does the forecast error decrease most strongly over the life of the contract?

1.5 Contribution and Structure of the Thesis

The main contribution lies in the competitor set. Existing evaluations of Kalshi macro contracts compare the prices with statistical models and professional surveys; here they are additionally set against gradient boosting machines and neural networks trained on the market's own hourly price paths. That is the demanding form of the accuracy question, since a model fitted to the price and beating it would show the price to be improvable by anyone willing to fit one.

The cross-variable design studies the three targets under one set of hypotheses and one evaluation protocol with the target-specific adaptations recorded, so that differences in performance can be attributed to the targets rather than to unrelated choices in three separate pipelines. That shifts the question from whether prediction markets forecast well in general towards when they add value.

A narrower contribution bears on how such prices are used: point accuracy and distributional calibration are tested and reported independently, rather than one being taken as evidence for the other.

The scope is correspondingly limited. The evidence comes from one platform over one sample period, from the contract families that platform lists, and from the benchmarks described above. No trading strategy is evaluated, no risk-premium correction is applied to either the contract prices or the futures benchmarks, and the model comparisons speak only to the architectures and training designs actually estimated. Chapter 6 states these constraints in full.

Chapter 2 reviews the literatures the analysis draws on, develops the conditions under which a contract price can be read as a probability, and specifies the benchmarks and the models. Chapter 3 covers the trade-level data, the conversion of a contract ladder into a distribution, and the sample audit trail. Chapter 4 sets out the evaluation design and reports H1, H2, H3 and H5 target by target, while Chapter 5 takes up H4 and H5c, which require all three targets at once. Chapter 6 states what limits the results, and Chapter 7 concludes.

Chapter 2

Literature, Theory and Methods

This chapter draws on four areas of research that support the thesis design. It begins with the conditions under which a prediction-market contract price can be interpreted as a probability. The following sections review market-based forecasts of economic variables, introduce the forecast-evaluation methods and discuss market efficiency as a basis for the equity results. The chapter then defines the benchmarks and models used to evaluate Kalshi.

2.1 Prices as Probability Forecasts

A prediction market rests on the premise that a price set by traders who stand to gain or lose money on an outcome embodies more information than any single trader holds. Wolfers and Zitzewitz (2004) provide the standard statement of this idea and the taxonomy associated with it. They distinguish between winner-take-all contracts, whose price tracks the probability of a single event, and index contracts, whose price tracks the expected value of a continuous outcome. Here, "index contract" does not specifically mean a contract about a stock-market index such as the S&P 500. It refers to a contract whose payoff changes with the numerical value of the outcome.

Index contracts are more relevant here, because a set of yes-or-no range contracts written on the same variable can be read as a step-wise cumulative distribution and converted into a mean and higher moments.

The evidence they survey, drawn largely from the Iowa Electronic Markets and from company-internal markets, indicates that contract prices frequently match or exceed the accuracy of polls and expert judgement. Kalshi's ladders are families of exactly this type, which is the warrant for the density construction of Chapter 3.

The analytical content of the aggregation claim belongs to Wolfers and Zitzewitz (2006). When traders have different beliefs and logarithmic utility, the equilibrium price equals the average of their beliefs. Under a broader range of risk preferences, the price generally remains close to this average, provided that traders do not disagree too strongly. This approximation is most reliable when prices are not close to zero or one. If traders' beliefs are imperfect but unbiased estimates of the true probability, averaging them reduces individual errors. The market price can therefore provide an efficient summary of the information held by different traders.

What a price delivers, though, is a risk-neutral probability, and a forecaster wants the physical one. The two separate whenever traders are not risk-neutral, since a risk-averse trader will pay more than the expected monetary payoff to insure against an outcome.

If we naively interpret the higher price as a higher physical probability, we overstate the event's real-world likelihood.

Manski (2006) shows that when traders hold heterogeneous beliefs, the price of a yes-or-no contract need not equal the average belief across traders, and may instead be a biased and compressed transformation of it. This result implies that the inference from price to probability depends on the behavioural model one is willing to assume.

Wolfers and Zitzewitz (2006) confirm Manski's conclusion. However, under plausible assumptions about risk attitudes and the distribution of disagreement, they show that the price remains close to the mean belief, so that the wedge Manski identifies is typically small. This debate is directly relevant, because the thesis treats Kalshi range-contract prices as risk-neutral probabilities without assuming the reading is safe. The calibration test reported in the empirical chapter, which is based on the probability integral transform, examines whether that inference holds closely enough for the forecasting comparison to be fair.

Two further results give more insights for the empirical work. Berg et al. (2008) evaluate the Iowa Electronic Markets across five presidential elections and find that market prices are closer to the eventual outcome than the polls in about three-quarters of the comparisons, with the advantage growing at horizons of more than one hundred days. Polls and surveys do not share this long-horizon advantage, which makes it relevant to the horizon analysis of Q4.

Snowberg et al. (2013), however, draw the line followed in this thesis: any advantage over competing methods must be demonstrated with data rather than inherited from the aggregation argument. The advantage can fail when traders share the same mistaken information, a small number of well-funded participants dominate, or risk premia distort prices.

2.2 Market-Based Forecasts of Economic Variables

Gürkaynak and Wolfers (2006) study a market that put the idea of using traded contracts to forecast economic releases to work two decades before Kalshi. Economic derivatives – financial contracts whose payoffs depended on upcoming economic announcements – wrote contracts on releases such as payrolls and retail sales. The authors find that the central forecasts implied by these contracts were similar to the average forecasts of professional surveys. At the same time, a family of related contracts provided information about the range and probability of possible outcomes.

Kitsul and Wright (2013) apply a similar method to CPI options and recover risk-neutral inflation distributions. Their results show that these distributions assigned more probability to high inflation and deflation than the observed frequency of those outcomes supported. These studies therefore provide both a justification and a warning for this thesis: a set of contracts on the same release can be used to construct a forecast distribution, but market pricing may distort the probabilities in its tails. H2 consequently evaluates the tails separately from the centre of the distribution.

The literature using Kalshi data is recent and can be classified by what each study does with the market prices. One group treats the price itself as the object of study.

Bürgi et al. (2025) analyse transaction-level records covering more than three hundred thousand contracts, and what matters here is the asymmetry they document: contracts priced low resolve as winners too rarely to break even, contracts priced high resolve as

winners somewhat more often than their price implies, and the distortion is stronger for traders who accept existing offers than for those who post them. Their finding is important for this thesis because the outer contracts in a ladder are often low-priced and thinly traded, making their contribution to the forecast distribution potentially unreliable.

Greene (2026) works at minute frequency on whether the shape of a contract's own price history predicts its subsequent price. Being a single-author industry working paper rather than peer-reviewed work, it is used here only for what it documents about microstructure: thin trading produces unchanged and stale last-trade prices, and prices bounded at zero and one generate mechanical effects near the boundary. Chapter 3 addresses these problems by recording the staleness of each strike-level price.

A second group of studies uses Kalshi prices as measures or predictors rather than directly evaluating their forecasting performance. Swanson et al. (2025) interpret price movements around Federal Reserve announcements as evidence of changes in macroeconomic expectations. Mohanty and Krishnamachari (2026) use daily changes in contract prices to predict cryptocurrency volatility. Glaze et al. (2026) study contracts concerning statements made during corporate earnings calls and find that the prices are well calibrated.

None of these studies makes the accuracy of the price itself its main research question. As Kalshi prices are increasingly used in applied research, it becomes more important to establish whether they provide reliable forecasts and probabilities.

Closest to the present exercise is Diercks et al. (2026), who evaluate Kalshi macro contracts directly as forecasts, building implied distributions for variables including inflation and the policy rate, setting them against statistical models and professional survey panels, and finding the market competitive and at times superior. This thesis adopts their construction for turning a contract family into a distribution.

Two questions remain unanswered. Their comparisons do not include flexible models trained on Kalshi's own price history, so they do not test whether such models can improve on the market price. They also examine each variable separately and therefore do not ask whether the value of prediction-market information differs across types of outcomes. The other studies reviewed in this section do not address these questions either.

2.3 Efficiency and the Limits of Forecastability

Two forms of market efficiency are relevant to this thesis and must be distinguished. The first concerns Kalshi. The market is informationally efficient if information available before the forecast cutoff cannot be used to improve on its price. This cannot be assumed from prediction-market theory and must instead be tested.

Q1 compares the Kalshi forecast with simple baselines, external benchmarks and trained models. Q2 examines whether market prices contain more useful information than the target's own history and whether history adds anything once the market prices are known.

The second form of efficiency concerns the underlying financial market and is relevant only to the S&P 500. Fama (1970) distinguish weak, semi-strong and strong market efficiency according to the information already reflected in prices. The weak form is most relevant here because past prices should not provide a reliable basis for predicting future returns. If this condition holds approximately, neither Kalshi nor a model using price history should consistently improve on the previous S&P 500 close by an economically

meaningful amount.

The previous close is therefore the main baseline for interpreting the equity results. This choice is also supported by Welch and Goyal (2008), who show that most proposed predictors of the equity premium fail to outperform a simple benchmark in out-of-sample tests.

Perfect market efficiency is not possible because information is costly to obtain. Grossman and Stiglitz (1980) argue that if prices already reflected all information completely, investors would have no incentive to collect and analyse it. Markets must therefore leave some opportunity for informed participants to recover the cost of research.

A small forecasting advantage based on timely information need not contradict market efficiency. Intraday order flow, for example, can reveal immediate buying and selling pressure that is absent from the previous official close. Such an advantage can be interpreted as compensation for producing information.

2.4 Evaluating Point Forecasts and Densities

A forecasting contest is only as credible as the test that decides it. Diebold and Mariano (1995) give the standard comparison: the test compares two forecasts through the difference in their losses and does not require either forecast to be correctly specified or the models to be nested. Because the original test can reject too often in small macroeconomic samples, this thesis uses the Harvey–Leybourne–Newbold correction supplied by Harvey et al. (1997). The corrected statistic accounts for small-sample bias and is evaluated using a Student's t-distribution. All pairwise accuracy tests in Chapter 4 use this correction.

Each hypothesis contains several comparisons. The Holm (1979) procedure is therefore applied separately within each hypothesis family to control the probability of false rejections.

The null hypothesis of the accuracy test is that the two forecasts have the same expected loss. Rejecting this null provides evidence that one forecast is more accurate. However, failing to reject it does not prove that the forecasts are equally accurate, because the sample may be too small to detect a genuine difference. Establishing practical similarity would require a prespecified margin defining the largest economically unimportant loss difference. This thesis does not define such a margin because there is no substantive basis for selecting one. H1b is therefore reported in three possible states: Kalshi is more accurate, Kalshi is less accurate, or the evidence is inconclusive.

An additional limitation applies to the S&P 500 comparisons. The corrected Diebold–Mariano statistic used in the implementation treats loss differences across events as serially uncorrelated. This assumption is relatively plausible for monthly CPI releases and infrequent Federal Reserve meetings, but less so for daily equity forecasts. Consecutive equity forecasts share information, and their errors may cluster during periods of high volatility.

If the loss differences are positively correlated, the current variance estimate is too small and the resulting daily p-values overstate the strength of the evidence. HAC standard errors or a block bootstrap that resamples consecutive days would be required for more reliable inference. Chapter 6 discusses this limitation further.

Point-forecast comparisons use only the central value implied by a contract ladder, although the ladder also provides a complete probability distribution. Diebold et al.

(1998) propose evaluating this distribution with the probability integral transform. For each event, the forecast distribution is evaluated at the realised outcome. Correctly specified distributions produce values that are uniform between zero and one and independent over time.

Concentration near zero and one indicates a distribution that is too narrow; concentration near the centre indicates one that is too wide; and a systematic shift indicates bias in the forecast centre. Independence additionally requires that forecast errors do not occur in persistent sequences.

The implementation tests uniformity but not serial independence. Uniformity is assessed using the Kolmogorov–Smirnov and Cramér–von Mises statistics, together with observed tail frequencies and central-interval coverage. The omission of an independence test is less important for macroeconomic events, which are separated by several weeks, but is a meaningful limitation for the daily S&P 500 series. The equity calibration result therefore concerns only the marginal distribution, not the temporal pattern of errors.

Because Kalshi ladders assign probabilities to a finite set of outcome ranges, the probability integral transform must be randomised. Rossi and Sekhposyan (2019) develop density specification tests that accommodate this while remaining valid when the density depends on estimated quantities. Calibration is not by itself a standard of quality, since a forecast that always reports the unconditional distribution is calibrated and useless, so Gneiting and Raftery (2007) pair it with sharpness: the distribution should be as concentrated as possible subject to being calibrated. The exact procedure is set out in Section 4.1.2.

Standard accuracy tests compare complete forecasts, whereas H3 asks which input set contains the useful information. The thesis therefore uses matched model pairs. Within each pair, the architecture and training design remain fixed while only the permitted inputs change. Differences in accuracy can thus be attributed to the input set rather than to model capacity or estimation method.

The pairs are evaluated with the same corrected loss-difference test, applied one-sided in the hypothesised direction. Holm corrections are conducted separately for H3a and H3b. These comparisons show which input set performs better under a given architecture, but they do not identify the mechanism or establish what another model specification might recover.

H4 compares performance across targets rather than within one target. Because CPI, the policy rate and the S&P 500 are measured in different units, Kalshi's error for each is expressed as a percentage of the corresponding benchmark's error. A paired bootstrap confidence interval is calculated for each ratio (Efron, 1979); the reference value of 100 represents equal error.

The procedure produces one interval for each target and does not directly test whether the ratios differ across targets. The ordering of the point estimates is therefore descriptive rather than a statistically established ranking, as Chapter 5 makes clear.

2.5 Prespecified Benchmarks

Forecast accuracy must be interpreted relative to an appropriate alternative, so H1 specifies the competitor set before estimation. Every benchmark may use only information available before the same cutoff as the Kalshi forecast. This prevents timing from giving a competitor an unfair advantage. The benchmarks also represent progressively richer

information, from persistence and estimated time-series patterns to institutional forecasts and conventional financial-market prices. Success against each benchmark therefore has a different interpretation.

The timing controls vary by target. Macroeconomic data require real-time vintages, official index closes do not undergo revision, and intraday market quotes require eligibility and freshness rules. Chapter 3 describes these procedures. The machine-learning models in Section 2.6 also form part of the H1 competitor set.

The no-change benchmark predicts that the next outcome will equal the most recently observed outcome. For the federal funds rate, it carries forward the target-range upper bound set at the previous meeting. This is a strong baseline because the Federal Reserve usually leaves the rate unchanged. Within the twelve-event test sample, it forecasts nine meetings correctly; the remaining three are the rate cuts. A competing forecast therefore adds most value when it anticipates a policy turning point.

For the S&P 500, the no-change rule uses the previous official close. It is difficult to beat because daily index movements are usually small relative to the index level. CPI instead uses an autoregressive model as its prespecified baseline. Monthly inflation is persistent but may also exhibit short-run dynamics across recent releases, which the AR(3) is intended to represent. Earlier research shows that simple benchmarks can be difficult to outperform for inflation and equity returns (Atkeson & Ohanian, 2001; Welch & Goyal, 2008).

The AR(3) benchmark predicts each target from its three most recent observations. It is estimated by least squares at every forecast origin using only the history available at that time. The model contains few parameters, making it suitable for the limited macroeconomic samples, and tests whether the target's recent path improves on simple persistence.

The three lags represent monthly releases for CPI, policy decisions for the federal funds rate and trading days for the S&P 500. For the index, the model is fitted to daily log returns and the predicted return is applied to the previous close. It therefore tests short-run return predictability rather than reproducing persistence in the index level.

Prices from conventional financial markets aggregate the expectations of financially motivated traders and are often interpreted as forecasts of variables affecting their payoffs (Fama, 1970; Gürkaynak & Wolfers, 2006). Unlike event contracts, however, these instruments do not settle directly on one economic release. Their prices must be converted into forecasts, and the conversion may retain risk and liquidity premia, particularly in the tails (Kitsul & Wright, 2013). H1b uses these benchmarks because they represent information already available from established institutions and markets.

For CPI, the external benchmark is the latest Cleveland Fed inflation nowcast published before the forecast cutoff (Knotek & Zaman, 2017). The nowcast is produced by the Cleveland Fed rather than estimated within this thesis.

The policy-rate benchmark uses the 30-day federal funds futures contract for the month after each meeting (Gürkaynak et al., 2007). Subtracting its settlement price from 100 gives the expected average effective federal funds rate for that month. Kalshi instead settles on the upper bound of the Federal Reserve's target range, so the two quantities are not directly comparable.

The futures-implied rate is converted by adding the latest observed difference between the target upper bound and the effective rate. This puts the forecasts in the same units but does not remove the futures risk premium (Piazzesi & Swanson, 2008). Since forecast differences are measured in basis points, the remaining premium limits the interpretation

of close comparisons; Chapter 6 discusses this issue. A futures settlement dated on the cutoff day is excluded to avoid using information that may have arrived after the forecast.

The S&P 500 benchmark is constructed from licensed intraday E-mini futures data. A futures quote cannot be used directly as a forecast of the cash index because futures and cash prices differ with financing costs, expected dividends and time to expiry. The continuation series does not provide a fixed expiry, so the required cost-of-carry adjustment cannot be calculated.

The implementation instead anchors the conversion on the latest observed futures–cash basis. It adds the futures change since the previous cash close to that closing level, assuming a constant basis between the anchor and forecast cutoff. Bars stamped exactly at the cutoff are excluded, and quotes more than 90 minutes old are rejected. Because the licensed data begin after the test period starts, the benchmark is available for 158 of 229 test days. All comparisons involving it are recalculated on this shorter common sample.

Table 2.1: Prespecified Forecast Benchmarks

Benchmark	Variable	Information at cut-off	Construction
No change	FFR, S&P 500	last outcome	none
AR(3)	all three	three own lags	least squares, refitted at each origin
Cleveland nowcast	CPI	daily institutional nowcast	none (published)
Fed funds futures	FFR	settlement before cutoff	EFFR-to-upper-bound basis conversion; no premium adjustment
E-mini future	S&P 500	prior cash close and futures bars	prior-close basis anchor; 90-minute freshness rule

Notes: Prespecified benchmarks by target variable, information source and construction.

Table 2.1 summarises a hierarchy of increasingly demanding comparisons. The no-change rule asks whether Kalshi contains information beyond the latest observation; the AR(3) asks whether it improves on recent time-series behaviour; and the external benchmark asks whether it adds to an established institutional forecast or financial-market instrument.

Only the external comparison measures Kalshi's marginal value relative to information already available elsewhere. It also uses the shortest samples and the strongest competitors. Results are therefore classified as Kalshi being more accurate, less accurate or statistically inconclusive, not as evidence of equivalence.

2.6 The Machine-Learning Models

H1c tests whether a flexible model using information available to the market can forecast more accurately than the Kalshi price. Six model configurations are estimated for each target. Gradient boosting is applied to the market, history and hybrid information sets; an MLP is applied to history; and an LSTM is applied to the ordered market and hybrid sequences. Each configuration is estimated under three training designs, producing eighteen runs per target.

This is not a complete crossing of three model types and three information sets, because some combinations do not match the data structure. A short vector of historical lags does not require a recurrent model, while a standard feedforward network does not preserve the temporal structure of a market-price sequence explicitly.

The three information sets determine what each model may observe. The Kalshi set contains hourly price paths for the three highest-volume strikes, recording the price, associated outcome level and time since the last trade. The history set is a short vector of the target's previous values and has no within-event sequence. The hybrid set combines the market sequence with those historical variables.

H3 compares the information sets through matched model pairs in which the architecture and training design remain fixed. Differences in accuracy can therefore be attributed more directly to the permitted inputs.

Gradient boosting constructs an additive ensemble of shallow regression trees. Each new tree is fitted to the errors remaining from the existing ensemble, so the combined model gradually improves its predictions (Friedman, 2001). This method can identify non-linear relationships and interactions without requiring them to be specified in advance. Shallow trees and a low learning rate also reduce overfitting risk in small samples.

The settings are fixed across targets and information sets: 300 trees, a maximum depth of 2 and a learning rate of 0.05. The depth restriction allows simple pairwise interactions while limiting patterns that would be difficult to estimate reliably from the small macroeconomic samples.

The recurrent model is used for Kalshi price sequences because their order may contain information beyond the final observed price. Each sequence passes through a single-layer LSTM (Hochreiter & Schmidhuber, 1997) with 32 hidden units. The final hidden state summarises the price path and is combined with any static historical features before passing through a 32-unit prediction layer and one output unit. Training minimises weighted mean squared error using the Adam optimiser (Kingma & Ba, 2015), a learning rate of 10^{-3}, a batch size of 256 and gradient clipping at 1.0.

The history-only neural model uses the same training settings but is an MLP with layers of 32, 16 and 1 unit. It is labelled as an MLP because the history input is a static vector of lags rather than a sequence suitable for an LSTM.

The training designs use different parts of each contract's price history. The terminal design uses only the final window before each training event, producing one row per event. The unweighted design gives equal importance to every available window. The weighted design also uses every window but discounts observations exponentially with their distance from the event.

The discount half-life is 240 hours for CPI and the policy rate and 12 hours for the S&P 500, reflecting the different contract lengths. This weighting turns the H5a horizon pattern into a modelling choice tested by H5b. Because the half-life was not theoretically derived or formally fixed before the horizon profiles were examined, H5b is treated as a robustness check rather than a confirmatory test of a prespecified value.

Early stopping uses a validation block formed from the most recent training events. Five events are reserved for CPI and the policy rate because their samples contain only a few dozen events. Twenty trading days are reserved for the S&P 500 to provide a less noisy stopping signal.

The maximum number of epochs differs by training design. The terminal design may train for 300 epochs with a patience of 30 because it contains one row per event. The full-window designs may train for 100 epochs with a patience of 10 because each event

contributes many more observations.

A typical CPI event provides about 190 windows, a policy-rate event about 240 and an equity event about 13. Macroeconomic contracts can trade for weeks, whereas equity contracts normally cover a day or two. Consequently, one event does not represent the same number of training rows across targets. For the tree models, terminal windows are represented through summary features, while full-window observations are supplied as flattened vectors.

The models are re-estimated as the test period progresses so that no forecast uses parameters fitted on information published after its cutoff. CPI and policy-rate models are refitted before each event.

Daily refitting is computationally impractical for the S&P 500. The equity models are therefore refitted at the beginning of each month and held fixed until the next refit. Their parameters may be nearly one month old for forecasts produced late in the month. This disadvantages the models relative to the continuously updating Kalshi market and makes the S&P 500 H1c result conditional on monthly refitting. Although the code includes a weekly option, all thesis results use the monthly cadence.

No hyperparameter is selected using the final test data. Settings are fixed before test evaluation, and early stopping uses only validation events from the training period. The models may therefore fall short of their best attainable accuracy, but their comparison with Kalshi is not created by repeated tuning on the evaluation sample.

All targets share the same six configurations, three training designs, loss function, rolling-origin structure and general stopping procedure. Numerical choices differ where the data require it, including validation-block size, discount half-life, refit frequency, horizon grid and windows per event. These adaptations are reported because identical settings for monthly releases, infrequent policy meetings and hundreds of trading days would not create a substantively comparable design.

Chapter 3

Data

All market data come from one source, a complete trade-level history downloaded through Kalshi's public trade interface and frozen in July 2026. The datasets are constructed using the same general method, described in Section 3.2. The later target-specific sections therefore discuss only the features that differ across CPI, the federal funds rate, and the S&P 500.

Table 3.1 summarises the data sources and their roles for the three targets.

3.1 Sources and Targets

Table 3.1: Data Sources

Data	Variable	Provider	Role
Contract trades	all three	Kalshi trade API	market forecasts, model inputs
Release timestamps	CPI	BLS release calendars	forecast cutoffs
Statement times	FFR	Federal Reserve calendar	forecast cutoffs
First-release CPI	CPI	FRED/ALFRED	labels, history
Target-range upper bound	FFR	FRED/ALFRED	labels, history
Official index closes	S&P 500	FRED	labels, history
Inflation nowcast	CPI	Cleveland Fed	external benchmark
Fed funds futures	FFR	LSEG	external benchmark
E-mini S&P 500 futures	S&P 500	LSEG	external benchmark

Notes: Data sources by target. All snapshots were frozen in July 2026 and archived in the reproducibility package.

Each target is defined as the number on which the Kalshi contracts settle, so that market, benchmarks and models are scored against the same quantity: the seasonally adjusted month-over-month change in the headline Consumer Price Index as first published and rounded to one decimal place, the upper limit of the federal funds target range set at each scheduled meeting, and the official daily closing level of the S&P 500. Defining the targets in this way ensures that Kalshi, the external benchmarks, and the statistical models are evaluated against the same realized quantity. The S&P 500 nevertheless

requires different timing rules because its value changes continuously while the prediction market is open, whereas CPI and FOMC outcomes are revealed through scheduled announcements.

3.2 Common Construction

3.2.1 From Contract Prices to a Distribution

Recovering a distribution from contracts on a common release is established in the literature. Gürkaynak and Wolfers (2006) apply the approach to digital options on macroeconomic releases, and Christoffersen et al. (2013) review density recovery from market prices and the required shape restrictions. The present implementation follows Diercks et al. (2026) in treating a Kalshi contract family as a continuously updated distribution rather than as a single probability: together, its prices describe the centre, spread and tails of the expected outcome. The implementation is adapted to the two contract structures in the sample. Threshold contracts, used for CPI and FFR, describe cumulative probabilities; between-bracket contracts, used mainly for the S&P 500, assign probability mass directly to an interval.

CPI and policy-rate contracts ask whether an outcome will exceed a threshold x. For ordered thresholds $x_1 < \cdots < x_K$ and final pre-cutoff prices $p_1, \ldots, p_K$, prices are clipped to the unit interval. A running minimum from the lowest strike imposes a non-increasing exceedance curve:

$$\tilde{S}(x_k) \;=\; \min_{j \le k} \min\bigl\{\max(p_j,\, 0),\, 1\bigr\}, \tag{3.1}$$

This step removes small cross-strike inconsistencies caused by stale or thin trading. It is the shape restriction identified by Christoffersen et al. (2013) for densities constructed from separately traded contracts. Differences in the repaired curve give probability masses between adjacent thresholds. Tail mass is placed half of the minimum strike spacing beyond the outer thresholds:

$$m_0 = 1 - \tilde{S}(x_1), \qquad m_k = \tilde{S}(x_k) - \tilde{S}(x_{k+1}), \qquad m_K = \tilde{S}(x_K), \tag{3.2}$$

The support points are $z_0 = x_1 - h$, $z_k = \frac{1}{2}(x_k + x_{k+1})$, and $z_K = x_K + h$, where $h = \frac{1}{2}\min_k(x_{k+1} - x_k)$. The implied mean of the discrete distribution is

$$\hat{\mu} \;=\; \frac{\sum_k m_k z_k}{\sum_k m_k}, \tag{3.3}$$

with normalisation accounting for any mass lost during clipping (Gürkaynak & Wolfers, 2006).

S&P 500 contracts instead cover "between" brackets with lower- and upper-tail contracts. Each bracket price is treated as its probability mass at the bracket midpoint, so no monotonicity repair is needed. Tail mass is placed half of the median bracket width beyond the outer floor and cap. All masses are then normalised before applying (3.3).

Using every listed strike gives the full-ladder implied mean, fixed in advance as the primary market forecast. The top-3 implied mean uses the three most heavily traded strikes and serves as a sensitivity measure and as the model input. Both measures are stored for all targets. H4 is the exception: it uses the top-3 measure for the S&P 500 after the full-ladder results were observed. Chapter 5 therefore treats H4 as descriptive.

3.2.2 Point-in-Time Correctness

Each forecast cutoff is the time at which the outcome becomes public. Only trades strictly before that instant enter the data, and predictors are lagged to prevent same-day information leakage.

CPI and policy-rate labels and historical predictors use first-release vintages from a real-time archive. If a real-time observation is missing, the latest earlier release is used and the substitution is flagged. Official S&P 500 closes are not revised, so no vintage archive is needed. Instead, equity predictors are lagged to the previous official close, and the futures benchmark follows the timing and freshness rules in Section 3.5.

3.2.3 Sequence Windows

The sequential data convert irregular trades into fixed-shape windows using a common procedure. Each event is placed on an hourly grid from its first trade to the cutoff. For each strike and hour, the most recent executed price is carried forward. Staleness is recorded because last-trade prices can become outdated in thin markets (Greene, 2026). For a strike last traded a hours before a grid point, the staleness feature is

$$\phi \;=\; \min\bigl\{\ln(1+a),\; 8\bigr\}, \tag{3.4}$$

and an untraded strike receives the sentinel value 8. The longest horizon is 1,440 hours and $\ln(1+1440) \approx 7.27$, so the sentinel exceeds any observed staleness and distinguishes no trade from an old trade.

Consecutive hourly windows are created at a fixed stride from the maximum horizon to the cutoff. The final pre-cutoff window is the terminal window used for point-forecast comparisons. Earlier windows support model training and the H5 horizon analysis.

At each window end, the three strikes with the highest cumulative volume are selected using only information available at that time. These contracts form three slots, ordered from the lowest to the highest strike so that their meanings remain stable. For each slot, the model receives the most recent price, the corresponding strike level, and the staleness of that price. Each timestep therefore contains $3 \times 3 = 9$ features:

$$(p_1, x_1, \phi_1,\; p_2, x_2, \phi_2,\; p_3, x_3, \phi_3), \tag{3.5}$$

where p_k, x_k, and ϕ_k denote the price, strike, and staleness of slot k. This fixed-width representation retains the probability, outcome level, and information freshness of the most actively traded contracts. If only two traded strikes are available, the final slot repeats the higher strike to preserve the nine-feature input. Windows with no traded slot are removed.

Only the top-3 representation is stored as a sequence tensor. A separate full-ladder sequence is not constructed because the number of listed contracts varies across events and over time. Instead, at each evaluation time, the latest available price for every listed contract is used to reconstruct the complete distribution and calculate one full-ladder implied mean. The terminal calculation uses prices strictly before the event cutoff; horizon calculations use prices available at the corresponding window end. The full-ladder mean is the primary market forecast, whereas the top-3 sequence is the fixed-dimensional input to the neural and tree models and also produces the top-3 sensitivity forecast.

For tree models, the same windows are flattened and joined to the same metadata.

No additional cleaning is applied. All targets share the feature layout, volume-based slot selection, staleness measure, and point-in-time rule. Window length, stride, and maximum horizon are scaled to the target's clock.

3.3 Consumer Price Index

Kalshi lists threshold contracts for each monthly CPI release, usually spaced by 0.1 percentage points. The raw file contains 168,904 trades for 66 events from July 2021 to July 2026. Kalshi changed the series code during the sample. The two generations were merged after confirming that no data month appeared under both codes.

The cutoff is the official release time from current and archived Bureau of Labor Statistics calendars, converted from Eastern Time with the applicable daylight-saving rule. In the collected sample, no CPI trades are timestamped at or after the corresponding release, however this rule is used in the implementation as Kalshi's documentation states that closure conditions are market-specific and that trading can occasionally continue while official confirmation is pending. The October 2025 all-items release was cancelled after a lapse in federal appropriations prevented data collection. Delayed releases retain their actual publication times.

Labels use the first published monthly index levels in ALFRED. Monthly changes are calculated from first-release levels and rounded to one decimal place. The same series provides the twelve most recent published monthly changes used as history predictors. The 2025 shutdown removes two labels: October was never published, and the November monthly change cannot be calculated without an October base. The November market remains in the trade data but is unlabelled.

The external benchmark is the Cleveland Fed daily nowcast of monthly headline CPI inflation (Knotek & Zaman, 2017). The official chart data contain 4,604 observations but no intraday timestamps. Each value is therefore treated as available at 10:00 Eastern Time on its business day, consistent with the stated update practice. The benchmark uses the latest target-month nowcast strictly before the cutoff. This timing assumption is a limitation.

The geometry in Table 3.3 produces 8,789 windows.

3.4 Federal Funds Rate

Policy-rate strikes lie on the 25-basis-point grid of possible outcomes. The market and label therefore share the same discrete space, and the implied mean can equal the realised value. The file contains 203,450 trades for 46 meetings from July 2021 to July 2026. Relisted contracts are mapped to the original series code, and one malformed ticker is standardised. Near a meeting, the ladder often concentrates most probability on one rate, producing near-degenerate terminal distributions. This pattern does not occur for CPI.

The cutoff is the 14:00 Eastern Time statement release on the decision day. The FOMC calendar is published in advance. No federal funds rate trade in the collected sample is timestamped at or after the corresponding release. The implementation nevertheless enforces the cutoff because Kalshi's documentation states that closure conditions are market-specific and that trading can occasionally continue while official confirmation is pending.

Labels are first-release values of the target-range upper bound, observed two days after each decision. The decision date is used rather than month-end to handle changes effective on the following day and meetings at month-end. The two-day lag ensures that the archive contains the new target. All calculations remain in upper-bound units because Kalshi contracts refer to that value. The series is not rounded or revised. The history vector contains the previous eight meeting outcomes, approximately one calendar year.

For the external comparison, the futures-implied average effective rate for the month following each meeting is converted to target-upper-bound units using the latest pre-cutoff basis described in Section 2.5. This is only a unit conversion. No adjustment is made for the predictable futures risk premium (Piazzesi & Swanson, 2008), which therefore remains a limitation of the policy-rate comparison in Chapter 6.

Window geometry matches CPI. Meetings occur six to seven weeks apart, so most retained events contribute the full set of about 240 windows. The final sample contains 8,391 windows for 35 events. Policy-rate ladders often drift slowly and then concentrate after data releases or committee communication, whereas CPI ladders usually remain dispersed until closer to publication.

3.5 S&P 500

S&P 500 events are trading days, so the sample contains many short-lived markets and horizons are measured in hours. Metadata show a median of 15 listed brackets and a mean of 19.5 per event; the median number traded is 10. The file contains 1,152,992 trades across 1,064 tickers. They merge into 1,051 date-level events from April 2022 to June 2026.

The index contracts were relisted under several series labels. Same-day markets are merged by calendar date, while listing generation remains a model feature because market depth and bracket rules changed. Duplicate brackets from overlapping generations are removed before feature construction. Bracket floors and caps come from market metadata, with ticker parsing used only when metadata are unavailable.

The cutoff is the scheduled close recorded in the metadata. It is usually 16:00 Eastern Time and earlier on shortened sessions. Labels are official FRED closing levels matched by date, with Kalshi settlement values used only for validation. The series has no publication lag or vintage revisions, as noted in Section 3.2.2.

Models use daily log returns rather than index levels because a fixed point error has a different scale at 4,000 and 6,000. This transformation is standard in equity forecasting (Rapach & Zhou, 2013). With S_t denoting the close on day t,

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right), \qquad \hat{\sigma}_t = \mathrm{sd}\big(r_{t-20}, \ldots, r_{t-1}\big), \qquad \ell_{k,t} = \ln\left(\frac{K_k}{S_{t-1}}\right), \tag{3.6}$$

where $\hat{\sigma}_t$ is twenty-day realised volatility and $\ell_{k,t}$ is the log-moneyness of slot k's representative strike K_k relative to the previous close. Moneyness makes contracts comparable across index levels (Christoffersen et al., 2013). History features are ten daily log-return lags, five- and twenty-day momentum, $\hat{\sigma}_t$, and the listing-generation indicator. All are lagged to the previous close. Predictions are converted back to levels for reporting, and results are stated in levels and returns.

The external benchmark is the E-mini S&P 500 future from licensed intraday data. Christoffersen et al. (2013) note that a futures quote observed after the cash close can

forecast a later spot price better than the stale close. The benchmark anchors on the previous official cash close and adds the futures movement since the last bar before that close. It uses the last completed bar strictly before the cutoff and rejects quotes more than ninety minutes old. Section 2.5 defines the calculation. The licensed record covers 158 of 229 test days, and all related comparisons use this common sample.

The full-ladder mean is the prespecified primary forecast and is calculated at the end of every window. The top-3 mean is a sensitivity measure and the model input. They differ more for the S&P 500 than for the macroeconomic targets. Chapter 4 reports the difference. The top-3 measure excludes much of a wide ladder, while the full ladder may include stale outer brackets for a continuously moving underlying. Chapter 4 reports both measures.

Training and testing sessions for models last about seventeen to twenty-four hours, producing the geometry in Table 3.3 and 11,450 windows for 859 events. Models are refitted monthly and held fixed until the next refit. Weekly refitting exists in the code, but all reported S&P 500 results use the monthly schedule.

3.6 Sample Construction and Comparison

Table 3.2 traces how the raw events become the final modelling samples. Table 3.3 then compares the resulting dataset designs across the three targets.

Table 3.2: Sample Construction

	CPI	FFR	S&P 500
Raw events in the trade file	66	46	1,051
below the trade threshold	−12	−4	−84
fewer than two distinct strikes	−1	–	–
no usable window	–	–	−103
unresolved at the data freeze	−1	–	−4
outside the sample period	−5	−7	–
release cancelled (October 2025)	−1	–	–
missing close time or settlement	–	–	−1
Retained events	46	35	859
no first-release label (November 2025)	−1	–	–
Labelled events used in modelling	45	35	859
of which test events	15	12	229

Notes: From raw Kalshi events to the final samples. The trade threshold is 100 for the S&P 500 and 300 for the macroeconomic targets, because a daily market concentrates its trading into one session.

Three filters in Table 3.2 affect the empirical design. Low-trade events cannot support the sequence representation. A single traded strike cannot define a distribution because (3.2) requires at least two thresholds. Unresolved events are removed because they lack labels at the data freeze. Retained CPI months run from October 2021 to May 2026; retained meetings run from March 2022 to June 2026 and cover tightening, plateau, and cutting periods; retained trading days run from April 2022 to June 2026. Events from January 2025 form each test set. Evaluation is rolling, and each test forecast uses only previously resolved events for estimation.

Table 3.3: Dataset Design Comparison

	CPI	FFR	S&P 500
Event type	monthly release	FOMC decision	trading-day close
Outcome	first-release MoM	target-range upper	official close
Outcome grid	0.1 pp steps	25 bp steps	near-continuous
Uncertainty type	measurement	committee decision	moving asset price
Contract format	thresholds	thresholds on grid	ranges plus tails
Distribution recovery	survival curve	survival curve	bracket masses
Cutoff	release instant	statement, 14:00 ET	scheduled close
Label vintages	first release	first release	none (never revised)
Raw / retained events	66 / 46	46 / 35	1,051 / 859
Test events	15	12	229
Window shape	24 steps, stride 6	24 steps, stride 6	12 steps, stride 2
Maximum horizon	60 days	60 days	48 hours
Windows (tensor rows)	8,789	8,391	11,450
History features	12 monthly lags	8 meeting lags	10 daily lags
Trade threshold	300	300	100
Model refit	per event	per event	monthly

Notes: Design comparison of the three event datasets.

Table 3.3 summarises the design. All targets share the trade source, point-in-time rule, volume-based slot selection, staleness feature in (3.4), and nine-feature timestep. Their clocks, contract formats, outcome types, and refit schedules differ. These differences are explicit parameters within a common construction.

Chapter 4

Empirical Results

This chapter reports out-of-sample results for point accuracy (H1), distributional calibration (H2), incremental information (H3), and horizon patterns (H5). H4 and H5c require direct comparison across all targets and are reported in Chapter 5.

Results are organised by hypothesis. This structure shows where findings are common across targets and where they differ. Test periods are held out in time, and no forecaster is tuned on test outcomes. The samples contain 15 CPI releases from January 2025 to May 2026, 12 FOMC decisions from January 2025 to June 2026, and 229 trading days over the same period.

4.1 Evaluation Rules and Reading Guide

4.1.1 Losses, Tests and Table Conventions

For N test events, accuracy is measured by root mean squared error and mean absolute error, which are standard scale-dependent measures (Hyndman & Koehler, 2006):

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}, \qquad \mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{y}_i - y_i\right|. \tag{4.1}$$

All hypotheses use squared loss. MAE is also reported because large individual errors can produce differences between the two measures.

Accuracy differences are evaluated using the event-level loss differential between a competitor and Kalshi (Diebold & Mariano, 1995):

$$d_i = e^2_{i,\mathrm{comp}} - e^2_{i,\mathrm{Kalshi}}, \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{Var}}(\bar{d})}}, \tag{4.2}$$

The statistic uses a t distribution with $N-1$ degrees of freedom after the small-sample correction in Harvey et al. (1997). Section 2.4 discusses its interpretation. A positive statistic favours Kalshi; a negative statistic favours the competitor. p-values are two-sided unless stated otherwise. The sequential correction in Holm (1979) is applied within each hypothesis family because families use different common samples. Failure to reject is classified as inconclusive, not as equal accuracy, because no non-inferiority margin is defined.

As a distribution-free check, each comparison includes a 95 percent bootstrap interval for the RMSE difference based on 10,000 event-level resamples (Efron, 1979). Any disagreement between the two procedures is reported.

Equity inference has an additional limitation. The implementation estimates $\widehat{\mathrm{Var}}(\bar{d})$ from the sample variance and does not adjust for serial dependence across 229 consecutive trading days. Because forecast errors cluster with volatility, daily p-values may be too small. This issue is most relevant for close comparisons.

Section 3.2.1 defines two market forecasts. The full-ladder mean is prespecified as primary and determines the main verdict. The top-3 mean is a sensitivity measure and the input provided to the models. For CPI, their correlation is 0.997 and their mean absolute difference is 0.013 percentage points. For the policy rate, they are similar except in April 2026, when they differ by 0.255 percentage points. They differ more consistently for the index, as discussed in Section 4.2.2.

Section 2.6 defines six model configurations under three training designs, for eighteen runs per target. Stochastic configurations are estimated with seeds 1–5 and averaged into one ensemble forecast. Seeds are not treated as additional observations. Section 4.2.3 reports all runs, while later chapter tables summarise the matched comparisons and leading cases.

4.1.2 Testing Calibration

H2 tests whether a ladder is calibrated as a distribution. If F_i is the forecast distribution for event i, correct calibration implies uniform values of $F_i(y_i)$ (Diebold et al., 1998). Because Kalshi outcomes are discrete, the randomised transform is used (Czado et al., 2009; Rossi & Sekhposyan, 2019):

$$u_i = F_i(y_i^-) + v_i \left[F_i(y_i) - F_i(y_i^-)\right], \qquad v_i \sim \mathrm{U}(0,1), \tag{4.3}$$

Here $F_i(y_i^-)$ is the probability strictly below the realised outcome. Uniformity is tested with Kolmogorov–Smirnov (Massey, 1951) and Cramér–von Mises (Anderson & Darling, 1952) statistics. Exact binomial tests cover the two 10 percent tails and central intervals, with Holm adjustment across seven tests. Calibration uses all labelled observations: 45 CPI releases, 35 meetings, and 859 trading days. Results are recomputed across 500 randomisation seeds.

Only marginal uniformity is tested. Serial independence, the second condition in Diebold et al. (1998), is not evaluated. This omission is especially important for 859 consecutive trading days, so H2 does not establish full dynamic calibration. The direction of any failure is also informative. Risk premia may be concentrated in extreme states (Kitsul & Wright, 2013), and low-priced contracts may be systematically overpriced (Bürgi et al., 2025). Both mechanisms produce priced tails that are too wide.

4.1.3 Matched Comparisons and the Horizon Grid

H3 uses matched model pairs that share architecture and training design and differ only in their inputs. Two model types across three designs produce six comparisons for each part of H3 and twelve per target. Reported p-values are one-sided in the hypothesised direction and Holm-adjusted within each part.

H5a uses fixed pre-event cutoffs. Each point reports forecast error and mean absolute revision from that horizon to the terminal forecast. The horizon grids are 6–480 hours for

CPI, 6–1,440 hours for policy decisions, and 2–36 hours for the index. Macroeconomic cutoffs use the same events throughout. The number of equity events decreases at longer horizons because not every daily contract is active 36 hours before the close.

H5b compares weighted and unweighted training with identical windows and model settings. A window observed h hours before its event receives weight

$$w(h) = 0.5^{h/H}, \tag{4.4}$$

which downweights training observations drawn from a different distribution than the evaluation point (Shimodaira, 2000). H is the target-specific half-life in Section 2.6.

4.2 Point Accuracy across Targets

4.2.1 Against the Time-Series Baselines (H1a)

Table 4.1: H1a: Accuracy against Prespecified Baselines

Target	Competitor	N	RMSE	Δ	95% bootstrap	DM	p
CPI (pp)	Kalshi, full ladder	15	0.0725	–	–	–	–
	AR(3)	15	0.2480	0.1756	[0.086, 0.252]	2.62	0.010
FFR (pp)	Kalshi, full ladder	12	0.0155	–	–	–	–
	No-change rule	12	0.1250	0.1095	[−0.013, 0.165]	1.88	0.043
	AR(3)	12	0.1496	0.1341	[0.081, 0.181]	2.75	0.019
S&P (index pts)	Kalshi, full ladder	229	46.51	–	–	–	–
	AR(3)	229	59.40	12.90	[3.60, 21.81]	2.81	0.005
	Previous close	229	59.62	13.11	[4.10, 22.15]	2.85	0.005

Notes: The full-ladder Kalshi mean is compared with each target's prespecified baseline on that target's full test sample. Δ is the baseline RMSE minus the Kalshi RMSE, so positive favours the market; the interval is a 95 percent event bootstrap of that difference. The p-column is one-sided and Holm-adjusted within the H1a family.

Table 4.1 shows that the market is more accurate than every prespecified baseline. For CPI, the AR(3) RMSE is 0.2480 percentage points, compared with 0.0725 for Kalshi. For the policy rate, the no-change rule is correct at nine meetings and misses the three cuts by 25 basis points. Its RMSE is 0.1250, about eight times the market's error.

The equity baselines are stronger. Under approximate market efficiency, the index should be close to a random walk (Fama, 1970). The previous close has RMSE 59.62 and the AR(3) has 59.40, so three return lags provide little additional accuracy. Kalshi improves on these values by about 22 percent, which is smaller than the improvement for either macroeconomic target.

For the policy-rate no-change rule, the Holm-adjusted one-sided p-value is 0.043, but the bootstrap interval [−0.013, 0.165] includes zero. The sample contains only twelve meetings, and three rate cuts account for the loss difference. The RMSE difference is large, but its precise magnitude remains uncertain.

Figure 4.1 identifies the largest CPI errors. Absolute errors exceed 0.10 percentage points in August 2025 (0.27 forecast, 0.40 realised), March 2025 (0.01, −0.10), January 2025 (0.29, 0.40), and June 2025 (0.20, 0.30). March 2026 has the largest realised value,

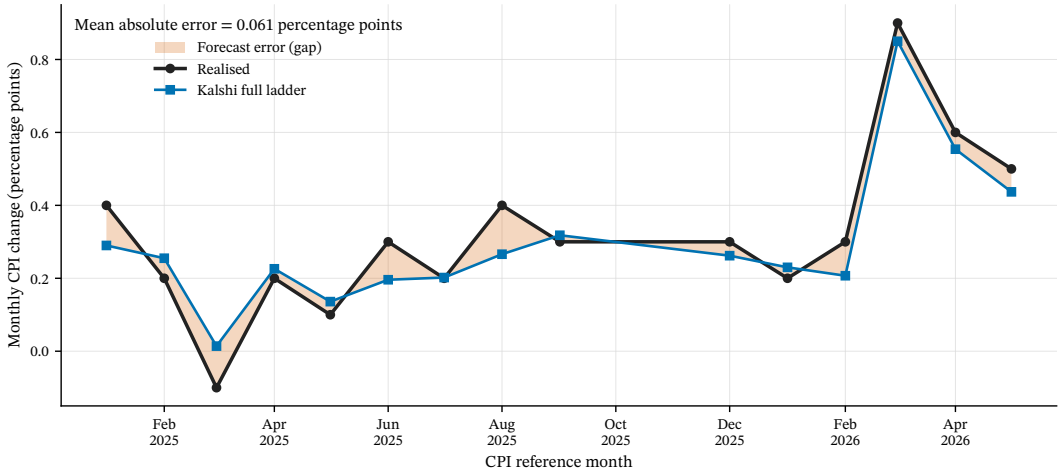


Figure 4.1: Terminal full-ladder Kalshi forecast and realised month-over-month CPI inflation, 15 test events. Shading is the forecast error.

0.90, and a forecast of 0.85. Three of the four largest errors are underforecasts. The mean signed error is −0.024 percentage points, consistent with the H2 asymmetry.

Policy-rate MAE is 0.012 percentage points, less than one twentieth of a 25-basis-point policy step. The largest error occurs in March 2025, with a forecast of 4.46 and outcome of 4.50. For the index, 10 of 229 errors exceed 100 points. The largest occurs on 28 November 2025, with a forecast of 6,621 and close of 6,849. Mean signed error is −12.1 points. The corresponding terminal full-ladder forecast-error plots for the policy rate and the index are reported in Appendix A, Figures A.1 and A.2.

H1a is supported for all targets on the primary reading, with adjusted p-values from 0.005 to 0.043. This comparison shows that the market improves on simple time-series baselines.

4.2.2 Against the External Benchmarks (H1b)

Table 4.2: H1b: Accuracy against External Benchmarks

Target	Benchmark	N	Kalshi	Bench.	Δ	DM	p
CPI (pp)	Cleveland nowcast	15	0.0725	0.0839	0.0114	0.88	0.394
FFR (pp)	Fed funds futures	12	0.0155	0.0146	−0.0009	−0.17	0.869
S&P (index pts)	E-mini futures	158	48.58	20.74	−27.85	−4.44	< 0.001
Sensitivity: the same comparison on the top-3 reading							
S&P (index pts)	E-mini futures	158	21.68	20.74	−0.94	−0.42	0.672

Notes: The Kalshi mean is compared with each target's external benchmark on the exact common sample. Δ is the benchmark RMSE minus the Kalshi RMSE, so positive favours the market. The final row is sensitivity evidence, not a second test of the hypothesis.

External benchmark results differ by target.
The CPI bootstrap interval is $[-0.012,\ 0.036]$.
For the policy rate, the interval is $[-0.012,\ 0.008]$.
The primary index interval is $[-37.24,\ -19.21]$.
For the top-3 reading, it is $[-5.20,\ 3.48]$.

For CPI, Kalshi RMSE is lower by 0.0114 percentage points, about 14 percent of the nowcast error, but the interval includes zero. The comparison is inconclusive. The Cleveland nowcast (Knotek & Zaman, 2017) remains much more accurate than the AR(3), with RMSE 0.0839 rather than 0.2480.

For the policy rate, federal funds futures have RMSE 0.0146 and Kalshi has 0.0155. The difference is 0.0009 percentage points, less than one tenth of a basis point, and cannot be resolved with twelve meetings. The established futures market (Gürkaynak et al., 2007) and Kalshi therefore produce similar point estimates, but the test does not establish equivalence. The futures risk premium remains unadjusted as described in Section 2.5.

For the index, H1b is rejected in favour of the futures benchmark. On 158 common days, futures RMSE is 20.74 and full-ladder RMSE is 48.58. The benchmark error is 57 percent lower, the bootstrap interval excludes zero, and the DM statistic is -4.44. Serial dependence described in Section 4.1.1 may affect the reported precision but is unlikely to reverse this large difference. The p-value should not be read at nominal precision. Table 4.2 reports the comparison.

The top-3 sensitivity measure has RMSE 21.68, compared with 20.74 for futures, and the difference is inconclusive. Across all 229 days, full-ladder and top-3 RMSE values are 46.51 and 24.98; median absolute errors are 18.8 and 7.0 points. The full ladder underpredicts the close by 12.1 points on average, while the top-3 mean overpredicts by 0.7 points. An index event has a median of ten traded brackets, and outer brackets may become stale as the underlying moves. Their inclusion can lower the full-ladder estimate in a rising market.

The full ladder remains the prespecified primary measure. The primary result is therefore that Kalshi is less accurate than the futures benchmark for the index. The top-3 result is sensitivity evidence that the aggregation rule contributes to the difference.

H1b is inconclusive for CPI and the policy rate and rejected in favour of the external benchmark for the index. The two inconclusive results do not demonstrate equal accuracy.

4.2.3 Against the Trained Models (H1c)

H1c asks whether any trained model attains lower out-of-sample error than the Kalshi implied mean. Each target has six model configurations under the terminal, unweighted and weighted training designs, producing eighteen runs. The stochastic forecasts are five-seed ensembles. Tables 4.3– 4.5 report every run alongside the market readings and benchmarks.

The Diebold–Mariano statistics compare each row with the full-ladder Kalshi mean. A positive statistic favours Kalshi and a negative statistic favours the forecast in that row. The displayed p-values are two-sided and unadjusted, allowing the individual comparisons to be read directly. The hypothesis verdicts in the text use Holm adjustment across each target's pooled competitor set.

Table 4.3: H1c: CPI Forecast Accuracy

Group	Forecast	RMSE	MAE	DM	p
Market	Kalshi, top 3	0.068	0.058	−1.27	0.224
	Kalshi, full ladder	0.072	0.061	–	–
External	Cleveland nowcast	0.084	0.072	0.88	0.394
Baseline	AR(3)	0.248	0.204	2.62	0.020
Terminal	GBM, Kalshi	0.131	0.106	2.73	0.016
	GBM, hybrid	0.135	0.105	1.66	0.119
	LSTM, Kalshi	0.145	0.124	3.18	0.007
	LSTM, hybrid	0.181	0.151	2.57	0.022
	MLP, history	0.226	0.164	1.75	0.102
	GBM, history	0.248	0.178	1.67	0.117
Unweighted	GBM, Kalshi	0.095	0.079	1.23	0.238
	LSTM, Kalshi	0.150	0.121	2.39	0.031
	LSTM, hybrid	0.154	0.124	2.31	0.037
	GBM, hybrid	0.200	0.138	1.41	0.182
	MLP, history	0.213	0.162	2.05	0.060
	GBM, history	0.247	0.167	1.58	0.137
Weighted	GBM, Kalshi	0.094	0.075	0.85	0.410
	LSTM, Kalshi	0.105	0.085	1.44	0.172
	LSTM, hybrid	0.149	0.124	2.56	0.023
	GBM, hybrid	0.173	0.129	1.65	0.121
	MLP, history	0.213	0.169	2.21	0.044
	GBM, history	0.244	0.175	1.66	0.120

Notes: Forecast comparison for CPI over the 15 test releases. Errors are in percentage points. DM statistics compare each forecast with the full-ladder Kalshi mean; p-values are two-sided and unadjusted.

Table 4.3 shows that no trained CPI model improves on either market reading. The full-ladder and top-3 RMSE values are 0.072 and 0.068, respectively. The best trained model is the weighted Kalshi-input GBM, with RMSE 0.094, about 30 percent above the full-ladder value. Its two-sided p-value is 0.410.

The Kalshi-input GBM improves from RMSE 0.131 under terminal training to 0.095 under unweighted training and 0.094 under weighted training. Although several individual rows have unadjusted p-values below 0.05, none of the eighteen model comparisons remains significant after pooled Holm adjustment. H1c is therefore supported for CPI: none of the tested models attains lower error than the market.

Table 4.4: H1c: Federal Funds Rate Forecast Accuracy

Group	Forecast	RMSE	MAE	DM	p
Market	Kalshi, full ladder	0.0155	0.0119	–	–
	Kalshi, top 3	0.0718	0.0260	0.96	0.358
External	Fed funds futures	0.0146	0.0100	−0.17	0.869
Baseline	No-change rule	0.1250	0.0625	1.88	0.087
	AR(3)	0.1496	0.1195	2.75	0.019
Terminal	GBM, history	0.2150	0.1360	1.95	0.077
	GBM, Kalshi	0.2635	0.1916	2.42	0.034
	GBM, hybrid	0.3124	0.2117	1.66	0.125
	LSTM, Kalshi	0.7989	0.5361	1.53	0.154
	LSTM, hybrid	1.3068	1.0104	2.22	0.048
	MLP, history	1.5256	1.3381	3.79	0.003
Unweighted	GBM, Kalshi	0.1954	0.1255	1.96	0.075
	LSTM, Kalshi	0.2051	0.1470	2.16	0.053
	GBM, history	0.2207	0.1368	1.85	0.092
	GBM, hybrid	0.2403	0.1602	2.08	0.062
	LSTM, hybrid	0.5285	0.3798	1.64	0.130
	MLP, history	1.0817	0.9648	4.32	0.001
Weighted	GBM, Kalshi	0.1640	0.1106	2.59	0.025
	GBM, history	0.2185	0.1354	1.88	0.087
	GBM, hybrid	0.2372	0.1520	2.01	0.070
	LSTM, Kalshi	0.2471	0.1635	1.34	0.206
	LSTM, hybrid	0.6686	0.4628	1.56	0.148
	MLP, history	0.7204	0.6477	3.76	0.003

Notes: Forecast comparison for the federal funds rate over the 12 test meetings. Errors are in percentage points. DM statistics compare each forecast with the full-ladder Kalshi mean; p-values are two-sided and unadjusted.

Table 4.4 shows a larger separation between the market and the trained models. The full-ladder market RMSE is 0.0155, while the best model, the weighted Kalshi-input GBM, has RMSE 0.1640. It therefore has more than ten times the error of the primary market reading.

The Kalshi-input GBM improves from 0.2635 under terminal training to 0.1954 under unweighted training and 0.1640 under weighted training, but remains far behind the market under every design. Six model rows are significantly worse before adjustment, but only one survives pooled Holm adjustment. For example, the weighted Kalshi-input LSTM has RMSE 0.2471 but an unadjusted p-value of 0.206. With only twelve meetings, a few large errors generate substantial variation in the loss differential. H1c is supported for the policy rate because no trained model improves on either market reading.

Table 4.5: H1c: S&P 500 Forecast Accuracy

Group	Forecast	RMSE	MAE	DM	p
Market	Kalshi, top 3	24.98	14.04	−4.20	< 0.001
	Kalshi, full ladder	46.51	31.61	–	–
External	S&P 500 futures (158 days)	20.74	14.58	−4.44	< 0.001
Baseline	AR(3)	59.40	45.56	2.81	0.005
	Previous close	59.62	45.57	2.85	0.005
Terminal	GBM, hybrid	26.85	18.47	−3.99	< 0.001
	LSTM, hybrid	26.98	18.66	−4.02	< 0.001
	GBM, Kalshi	27.40	18.73	−3.86	< 0.001
	LSTM, Kalshi	27.98	18.71	−3.74	< 0.001
	MLP, history	60.46	46.30	2.99	0.003
	GBM, history	62.37	47.74	3.09	0.002
Unweighted	LSTM, Kalshi	30.09	20.72	−3.42	0.001
	LSTM, hybrid	32.05	23.09	−3.14	0.002
	GBM, Kalshi	33.29	24.15	−2.86	0.005
	GBM, hybrid	43.83	30.30	−0.56	0.575
	MLP, history	60.29	46.67	3.02	0.003
	GBM, history	63.69	47.72	3.21	0.001
Weighted	LSTM, Kalshi	27.64	18.74	−3.84	< 0.001
	LSTM, hybrid	30.66	21.73	−3.36	0.001
	GBM, Kalshi	31.83	22.48	−3.12	0.002
	GBM, hybrid	38.88	26.57	−1.63	0.104
	MLP, history	60.50	46.62	3.04	0.003
	GBM, history	63.90	47.77	3.21	0.002

Notes: Forecast comparison for the S&P 500 over the 229 test days. Errors are in index points. The futures row uses the 158 futures-covered days and is not directly comparable with the remaining RMSE and MAE rows. DM statistics compare each forecast with the full-ladder Kalshi mean on the relevant common sample; p-values are two-sided and unadjusted.

Table 4.5 produces a different conclusion. Twelve of the eighteen trained models have lower RMSE than the primary full-ladder mean, and ten of these improvements remain significant after pooled Holm adjustment. The best run is the terminal hybrid GBM, with RMSE 26.85 compared with 46.51 for the full ladder.

All twelve models that improve on the full ladder use Kalshi price-path inputs. The six history-only models have RMSE values between 60.29 and 63.90 and are significantly worse. Market prices therefore contain the information that allows the trained models to improve on the full-ladder forecast.

No model improves on the top-3 RMSE of 24.98. The best trained model has approximately 7 percent higher error. Models can improve on a full-ladder mean affected by stale outer brackets, but not on the three high-volume brackets that form their own market input. The index models are refitted monthly, so their parameters can be almost one month old; this feature favours the continuously updating market. Three of the four best runs use terminal training, suggesting that excluding distant windows is more useful than retaining the full intraday history for this target.

Table 4.6: H1c: S&P 500 Forecast Accuracy in Log Returns

Group	Forecast	RMSE (%)	MAE (%)	DM	p
Market	Kalshi, top 3	0.401	0.220	−3.81	< 0.001
	Kalshi, full ladder	0.719	0.484	–	–
External	S&P 500 futures (158 days)	0.298	0.210	−4.40	< 0.001
Baseline	AR(3)	0.913	0.697	2.67	0.008
	Previous close	0.919	0.698	2.75	0.006
Terminal	GBM, hybrid	0.432	0.288	−3.62	< 0.001
Unweighted	LSTM, Kalshi	0.477	0.320	−3.12	0.002
Weighted	LSTM, Kalshi	0.444	0.292	−3.49	0.001

Notes: Selected forecast comparison for the S&P 500 in log-return units. Errors are multiplied by 100 and can be read as approximate percentage returns. The table shows both market readings, the external benchmark, the prespecified baselines, and the lowest-RMSE trained model from each design. The futures row uses 158 common days; all other rows use 229 days. DM statistics compare each forecast with the full-ladder Kalshi mean on the relevant sample; p-values are two-sided and unadjusted.

Table 4.6 shows that expressing errors as returns does not change the ranking. Full-ladder RMSE is 0.719 percent, compared with 0.919 percent for the previous-close rule and 0.913 percent for AR(3). The top-3 reading has RMSE 0.401 percent. The best model under each training design has lower RMSE than the full ladder, between 0.432 and 0.477 percent, with two-sided unadjusted p-values no greater than 0.002. None improves on the top-3 reading by raw RMSE. On the 158 days covered by the futures series, futures have the lowest displayed RMSE at 0.298 percent. The return-space evidence therefore confirms that the gains in Table 4.5 are not caused solely by changes in the nominal index level.

H1c is therefore supported for CPI and the policy rate and rejected for the S&P 500 on the prespecified full-ladder reading. This conclusion is restricted to the six model configurations, three training designs and stated S&P 500 refit cadence.

4.3 Distributional Calibration across Targets

Table 4.7: H2: Distributional Calibration

		Uniformity		Tail share		Central coverage		
Target	N	KS	CvM	Lower	Upper	50%	80%	90%
CPI	45	0.332†	2.186†	0.022	0.289†	0.489	0.689	0.778†
FFR	35	0.094	0.041	0.114	0.057	1.000†	1.000†	1.000
S&P 500	859	0.067†	1.224†	0.056†	0.056†	0.863†	0.952†	0.983†

Notes: Calibration of the terminal implied distribution for all labelled events. The S&P 500 ladder is read at the two-hour pre-close window. Tail shares have an expected value of 0.10 and coverage rates 0.50, 0.80 and 0.90. Holm adjusts across the seven tests within a target; † marks a rejection at the 5 percent level after adjustment.

H2 is rejected for CPI and the S&P 500. Uniformity and tail tests do not reject for the policy rate. Table 4.7 shows that the CPI and equity failures have opposite directions.

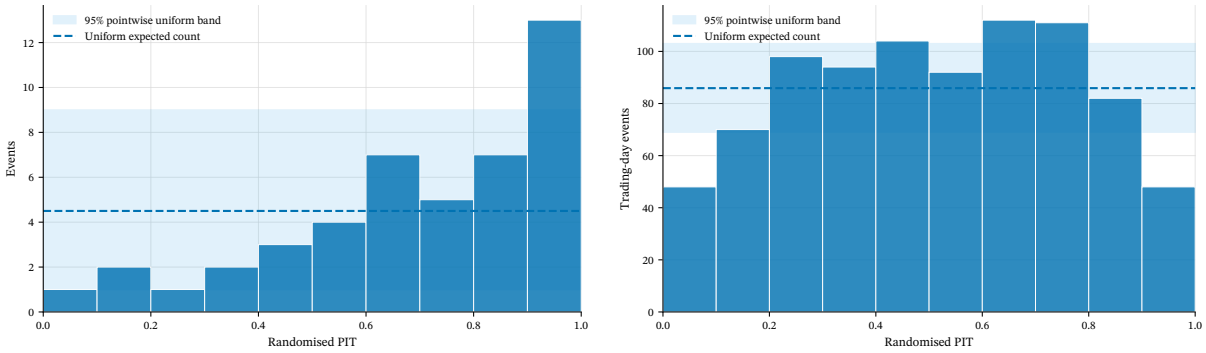


Figure 4.2: Randomised PIT values: 45 CPI releases (left) and 859 trading days (right). The dashed line is the count a calibrated ladder would produce and the band is a 95 percent pointwise interval around it.

The CPI ladder is too narrow in the upper tail. The share above the implied 90th percentile is 28.9 percent rather than 10 percent, while the lower-tail share is 2.2 percent. The 90 percent interval covers 77.8 percent of events, but 50 percent coverage is close to nominal at 48.9 percent. In the left panel of Figure 4.2, the final bin contains 13 of 45 observations. Risk premia and longshot overpricing would widen priced tails, while this result shows an upper tail that is too narrow, as discussed in Section 4.1.2. Uniformity is rejected for all 500 randomisation seeds, and the upper-tail share ranges from 0.178 to 0.400.

The index distribution is too wide. Each tail contains 5.6 percent of closes rather than 10 percent. Coverage is 86.3, 95.2, and 98.3 percent for nominal 50, 80, and 90 percent intervals. The right panel of Figure 4.2 is centred and hump-shaped, with mean PIT 0.510 and 48 observations in each end bin. Risk premia and longshot overpricing are consistent with this direction because they add probability to distant brackets. Uniformity is rejected for all 500 seeds, with tail shares between 0.040 and 0.078. The KS statistic is 0.067 and is precisely estimated because the sample contains 859 events; the large coverage excesses show the economic magnitude.

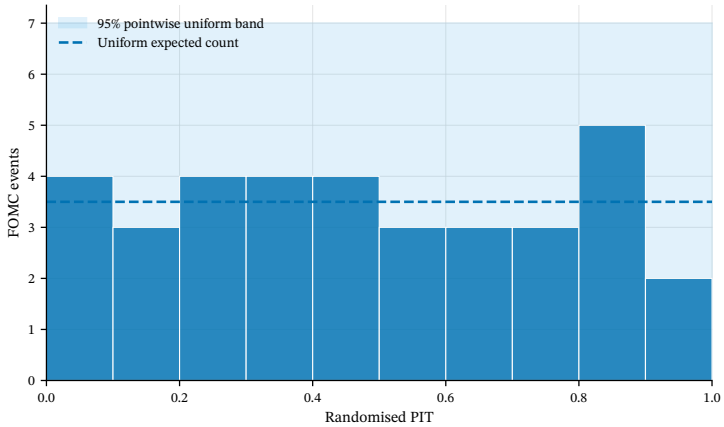


Figure 4.3: Randomised PIT values for 35 policy-rate decisions. The dashed line is the expected count under a uniform distribution.

The policy-rate case is affected by discreteness. Figure 4.3 shows that the PIT histogram appears approximately uniform, but this result requires caution. Uniformity and tail tests do not reject, while all three central intervals have 100 percent coverage. The

distribution assigns a mean probability of 0.913 and a median probability of 0.95 to the realised outcome; 30 of 35 distributions assign it more than 0.90. September 2024 is the main exception, when a 50-basis-point cut had probability 0.30. When one outcome has most of the mass, randomisation spreads the PIT over much of the unit interval and reduces test power. Central intervals also collapse to the dominant outcome, so their 100 percent coverage does not imply excessive width. Across 500 seeds, the KS test rejects 19 times and the CvM test 26 times. Non-rejection is common but not universal.

H2 therefore fails for CPI and the index. It is not rejected for policy-rate uniformity or tails, but the tests are weakly informative and do not establish calibration. All results concern marginal distributions because serial independence is untested. Point accuracy and calibration remain distinct: the CPI centre is accurate but its upper tail is too narrow, while the index distribution is too wide.

4.4 Incremental Information across Targets

The information-set results are reported in two stages. Table 4.8 summarises the direction and significance of the comparisons across targets, while Table 4.9 reports the underlying matched RMSE pairs.

Table 4.8: H3: Information-Set Results

	Target	Predicted direction	Bootstrap excludes 0	Raw $p < 0.05$	Survives Holm
H3a	CPI	6	3	1	0
	FFR	5	3	3	2
	S&P 500	6	6	6	6
H3b	CPI	6	2	1	0
	FFR	6	2	1	0
	S&P 500	4	3	3	3

Notes: Matched comparisons between information sets for H3a and H3b, six per hypothesis per target. Two models enter a pair only if they share a model type and a training design. Counts are out of six; "Holm" applies within each hypothesis and target.

Table 4.8 shows that the predicted H3a direction holds in 17 of 18 matched pairs and that the predicted H3b direction holds in 16 of 18 pairs. Holm-adjusted support is strongest for the S&P 500, while fewer comparisons survive adjustment for the two macroeconomic targets.

Table 4.9: H3: Matched Information-Set Comparisons

Model	Design	Market	Other	Δ	95% bootstrap
CPI — H3a: market against history only					
GBM	Terminal	0.1308	0.2478	0.1170	[−0.013, 0.254]
	Unweighted	0.0947	0.2470	0.1523	[0.011, 0.278]
	Weighted	0.0935	0.2443	0.1508	[0.031, 0.257]
Neural	Terminal	0.1446	0.2259	0.0814	[−0.022, 0.174]
	Unweighted	0.1501	0.2134	0.0634	[−0.025, 0.140]
	Weighted	0.1055	0.2129	0.1074	[0.042, 0.158]
CPI — H3b: market against hybrid					
GBM	Terminal	0.1308	0.1345	0.0037	[−0.049, 0.048]
	Unweighted	0.0947	0.2000	0.1053	[−0.006, 0.211]
	Weighted	0.0935	0.1726	0.0791	[0.004, 0.144]
Neural	Terminal	0.1446	0.1814	0.0368	[−0.011, 0.081]
	Unweighted	0.1501	0.1540	0.0040	[−0.027, 0.032]
	Weighted	0.1055	0.1488	0.0433	[0.001, 0.084]
FFR — H3a: market against history only					
GBM	Terminal	0.2635	0.2150	−0.0486	[−0.189, 0.089]
	Unweighted	0.1954	0.2207	0.0253	[−0.132, 0.180]
	Weighted	0.1640	0.2185	0.0544	[−0.071, 0.184]
Neural	Terminal	0.7989	1.5256	0.7267	[0.121, 1.407]
	Unweighted	0.2051	1.0817	0.8766	[0.578, 1.140]
	Weighted	0.2471	0.7204	0.4733	[0.196, 0.753]
FFR — H3b: market against hybrid					
GBM	Terminal	0.2635	0.3124	0.0489	[−0.122, 0.246]
	Unweighted	0.1954	0.2403	0.0449	[−0.114, 0.195]
	Weighted	0.1640	0.2372	0.0732	[−0.061, 0.199]
Neural	Terminal	0.7989	1.3068	0.5078	[0.291, 0.777]
	Unweighted	0.2051	0.5285	0.3234	[−0.006, 0.633]
	Weighted	0.2471	0.6686	0.4215	[0.172, 0.642]
S&P 500 — H3a: market against history only					
GBM	Terminal	27.40	62.37	34.96	[27.63, 42.46]
	Unweighted	33.29	63.69	30.40	[23.96, 37.24]
	Weighted	31.83	63.90	32.06	[25.63, 39.25]
Neural	Terminal	27.98	60.46	32.48	[25.85, 39.28]
	Unweighted	30.09	60.29	30.19	[24.60, 35.72]
	Weighted	27.64	60.50	32.85	[27.03, 38.88]
S&P 500 — H3b: market against hybrid					
GBM	Terminal	27.40	26.85	−0.55	[−3.15, 1.84]
	Unweighted	33.29	43.83	10.54	[5.70, 15.94]
	Weighted	31.83	38.88	7.04	[3.29, 11.49]
Neural	Terminal	27.98	26.98	−1.00	[−2.69, 0.77]
	Unweighted	30.09	32.05	1.95	[−0.48, 4.55]
	Weighted	27.64	30.66	3.02	[0.91, 5.40]

Notes: The 36 matched pairs behind Table 4.8. "Market" is the Kalshi-input model and "other" its history-only counterpart under H3a or its hybrid counterpart under H3b. Δ is other minus market, so positive supports the hypothesis. Units are percentage points for the CPI and FFR and index points for the S&P 500; intervals are 95 percent event bootstraps.

H3a compares market-input and history-only models. Table 4.9 reports the pairs summarised in Table 4.8. The exception to the predicted direction is the terminal policy-rate GBM, where market-input RMSE is 0.2635 and history-only RMSE is 0.2150. This pair has at most 34 training rows, and the policy rate is a step function that is often easy to extrapolate over short periods. Across designs, the history-only GBM has RMSE 0.2150–0.2207, while the simpler AR(3) reaches 0.1496.

For the index, market-input models have RMSE 27.64–33.29, compared with 60.29–63.90 for history-only models. All six pairs survive Holm adjustment with p below 0.001. For CPI, the weighted comparison is 0.0935 against 0.2443, but only one pair is significant before adjustment and none after Holm correction. For the policy rate, two neural pairs survive, with $p = 0.006$ and $p = 0.036$. The history MLP has RMSE 0.7204–1.5256, while all GBM intervals include zero and differences range from -0.0486 to 0.0544.

H3b tests whether target history improves a model that already uses market prices. For CPI and the policy rate, all twelve hybrid models are worse than their market-only counterparts. Under full-window training, CPI hybrid GBM RMSE rises from 0.0947 to 0.2000 and from 0.0935 to 0.1726. Policy-rate hybrid LSTM RMSE rises from 0.2051 to 0.5285 and from 0.2471 to 0.6686. Additional lags increase model dimension without providing enough independent events.

For the index, market-only models win all four full-window comparisons and three survive Holm adjustment. The hybrid GBM loses 10.54 points under unweighted training and 7.04 under weighted training. Under terminal training, the hybrid GBM and LSTM improve by 0.55 and 1.00 points, but neither difference is significant; their adjusted p-values are 1.000 and both bootstrap intervals include zero. Repeated history features add no new information across windows within the same day.

H3a and H3b are supported in direction for all targets, but inferential strength differs with sample size. Both are significant after adjustment for the index. For the policy rate, only the neural H3a pairs survive. No CPI pair survives multiplicity correction despite large RMSE differences. Overall, the price paths contain more useful information than target history for the tested models.

4.5 Horizon Evidence and Model Weighting

4.5.1 Error and Revisions over the Contract's Life (H5a)

Table 4.10: H5a: Forecast Accuracy by Horizon

CPI, hours before release									
	480	240	168	96	48	24	6		
RMSE	0.0952	0.0851	0.0888	0.0826	0.0827	0.0818	0.0720		
Revision	0.0368	0.0220	0.0227	0.0227	0.0161	0.0155	0.0108		
FFR, hours before statement									
	1440	720	480	240	168	96	48	24	6
RMSE	0.1278	0.0537	0.0508	0.0225	0.0228	0.0173	0.0172	0.0163	0.0157
Revision	0.0908	0.0387	0.0242	0.0073	0.0069	0.0038	0.0033	0.0021	0.0025
S&P 500, hours before close									
	36	24	12	6	4	2			
RMSE	98.69	99.51	95.79	69.74	59.60	51.86			
Events	101	106	121	214	226	229			

Notes: Full-ladder RMSE and mean absolute revision from the terminal forecast at fixed pre-event cutoffs. Units are percentage points for the first two targets and index points for the third. CPI and FFR cutoffs rest on 15 and 12 events throughout; the S&P 500 event count is given in the table because it falls with the horizon.

Table 4.10 shows that forecast errors and revisions generally decline as each event approaches, but none of the profiles is strictly monotonic. CPI RMSE rises from 0.0851 at 240 hours to 0.0888 at 168 hours and from 0.0826 at 96 hours to 0.0827 at 48 hours. Policy-rate RMSE rises from 0.0225 at 240 hours to 0.0228 at 168 hours. Index RMSE rises from 98.69 at 36 hours to 99.51 at 24 hours. H5a concerns the overall profile and is supported for all targets.

The profiles differ in magnitude. Policy-rate RMSE falls from 0.1278 to 0.0157, mainly between 60 and 10 days before the statement. CPI RMSE changes less, with the final value about one quarter below the starting error. Index RMSE falls by about half within 36 hours. Revision measures show the same ordering without using outcomes: policy-rate revisions decline from 0.0908 to 0.0025 percentage points, CPI revisions from 0.0368 to 0.0108, and index revisions from 48.9 to 11.9 points on the futures-common sample.

For the index, the event count declines from 229 at two hours to 101 at 36 hours. Longer-horizon observations therefore cover only contracts that opened early. The revision values and the following figure use the 158-day futures-common sample, whereas the RMSE series above covers up to 229 days. Chapter 5 uses the 229-day values for H5c.

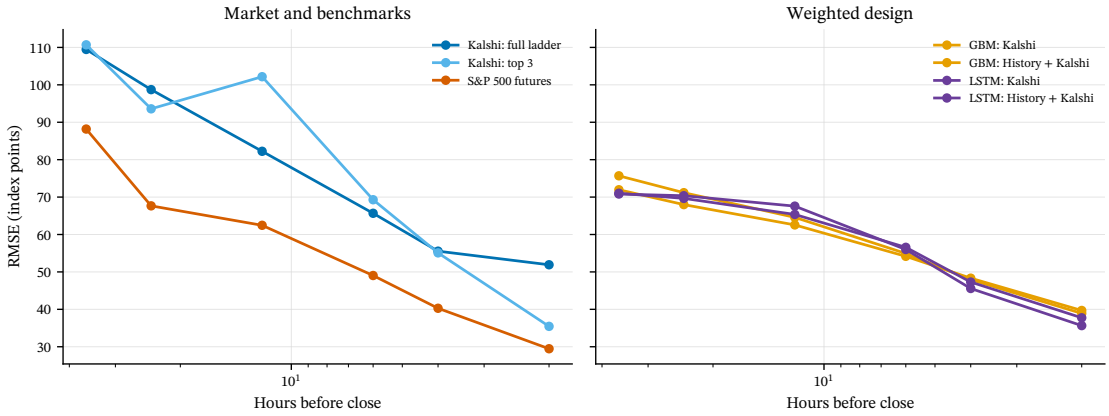


Figure 4.4: S&P 500 RMSE by forecast horizon on the 158-day futures-common sample: market readings and the futures benchmark (left), weighted-design models (right). Event counts fall from 158 at two hours to 41 at 36.

Figure 4.4 extends H1 across the contract horizon. All four model curves remain below the full-ladder curve at every cutoff and above the futures curve except at 36 hours. The full ladder is more accurate at 6, 12, and 36 hours, while the top-3 measure is more accurate within the final four hours by 16.5 points. This pattern is consistent with Section 4.2.2: late-session volume makes the most active brackets more current.

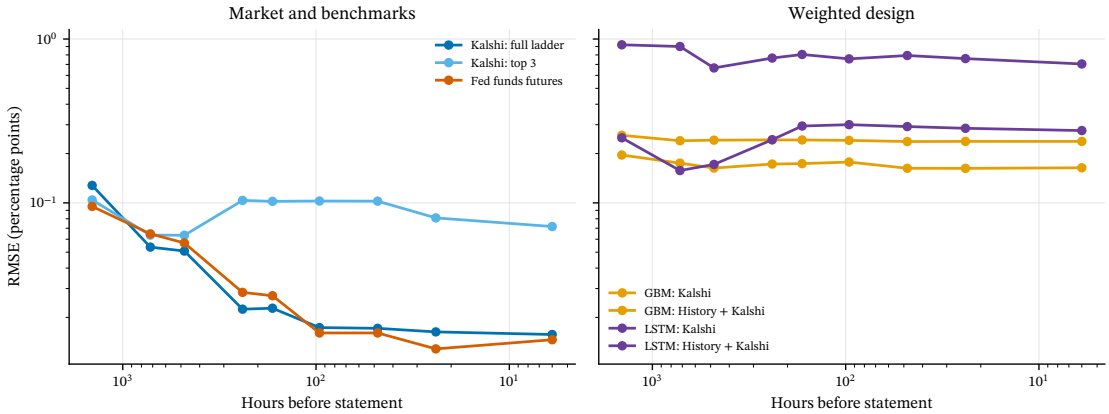


Figure 4.5: Federal funds rate RMSE by forecast horizon, 12 test meetings, on a logarithmic vertical scale: market readings and the futures benchmark (left), weighted-design models (right).

Figure 4.5 shows a different policy-rate pattern. Weighted model RMSE remains between 0.16 and 0.92 percentage points at all horizons, about an order of magnitude above the market readings, and does not decline near the statement. The futures curve remains close to Kalshi. At 1,440 hours, futures RMSE is 0.0952 and Kalshi RMSE is 0.1278; futures is slightly worse between 168 and 720 hours. The top-3 curve reaches 0.0634 at 480 hours, rises to about 0.1035 between 240 and 48 hours, and ends at 0.0717. The April 2026 meeting explains much of this irregularity.

H5a is descriptive evidence of an overall decline for all targets. Forecast error and remaining repricing decrease together as information arrives.

4.5.2 Weighted against Unweighted Training (H5b)

Table 4.11: H5b: Weighted versus Unweighted Training

	CPI		FFR		S&P 500	
Configuration	Unwt.	Wt.	Unwt.	Wt.	Unwt.	Wt.
GBM, Kalshi	0.0947	0.0935	0.1954	0.1640	33.29	31.83
LSTM, Kalshi	0.1501	0.1055	0.2051	0.2471	30.09	27.64
GBM, hybrid	0.2000	0.1726	0.2403	0.2372	43.83	38.88
LSTM, hybrid	0.1540	0.1488	0.5285	0.6686	32.05	30.66
GBM, history	0.2470	0.2443	0.2207	0.2185	63.69	63.90
MLP, history	0.2134	0.2129	1.0817	0.7204	60.29	60.50
Supporting H5b	6/6		4/6		4/6	

Notes: Terminal RMSE under the unweighted and weighted designs by configuration. Units are percentage points for the CPI and FFR and index points for the S&P 500. A configuration supports H5b when the weighted value is lower.

As Table 4.11 shows, weighting reduces terminal error in 14 of 18 configurations: 6 of 6 for CPI, 4 of 6 for the policy rate, and 4 of 6 for the index. H5b is fully supported for CPI and partly supported for the other targets.

For the policy rate, weighting worsens the Kalshi-input and hybrid LSTMs. The weighted Kalshi-input GBM is stable across seeds, with RMSE 0.16403–0.16404. The Kalshi-input LSTM ranges from 0.244 to 0.392 under weighting and from 0.229 to 0.492 without weighting. These overlapping ranges show that the LSTM ordering is within seed variation and cannot be resolved with twelve meetings.

For the index, weighting worsens only the two history-only configurations, by less than 0.5 percent of their error. Every model using a market path improves; the hybrid GBM declines from 43.83 to 38.88. However, terminal training is more accurate than weighted training for five of six index configurations. For a single-session contract, discarding distant windows may be more effective than downweighting them.

H5b is a robustness test of the model design. Improvement in 14 of 18 cases is evidence that the H5a horizon pattern is not solely an artefact of the evaluation grid.

4.6 Summary of Results

Table 4.12: Summary of Hypothesis Results

	CPI	FFR	S&P 500
H1a baselines	Supported	Supported	Supported
H1b benchmark	Inconclusive	Inconclusive	Rejected
H1c models	Supported	Supported	Rejected
H2 calibration	Rejected	Not rejected*	Rejected
H3a market vs history	Direction 6/6	Direction 5/6	Supported, Holm 6/6
H3b hybrid vs market	Direction 6/6	Direction 6/6	Direction 4/6
H5a horizon decline	Supported	Supported	Supported
H5b weighting	6/6	4/6	4/6

Notes: Verdicts on the prespecified reading by hypothesis and target. "Supported" refers to the prediction made by the hypothesis and does not always represent a direct Kalshi–benchmark comparison. "Direction $x/6$" means that x of six matched RMSE comparisons have the hypothesised sign; "Holm $x/6$" means that x also survive Holm adjustment. "Inconclusive" means that the accuracy difference cannot be resolved and no non-inferiority margin was tested. $*$ FFR uniformity and tail tests do not reject, but interval coverage does, and the near-degenerate ladder limits test power. No H2 verdict covers serial independence, which was not tested.

Table 4.12 condenses the hypothesis-level evidence. The point-accuracy rows use different comparisons and therefore require different interpretations. H1a favours Kalshi against each prespecified simple baseline. H1b is inconclusive for CPI and the policy rate and favours E-mini futures over the primary S&P 500 full-ladder reading; non-rejection in the first two cases does not establish equal accuracy. H1c finds no model with lower RMSE than the full ladder for either macroeconomic target. Twelve of the 18 index models have lower RMSE than the full ladder; the top-3 sensitivity reading remains more accurate than every model.

The calibration results qualify these point forecasts. CPI outcomes occur too often in the upper tail, whereas the index ladder is too wide. Policy-rate uniformity and tail tests do not reject, but the concentrated ladder leaves those tests with limited power and produces excessive interval coverage. These are findings about marginal calibration only because PIT independence is not tested.

The remaining rows describe patterns across matched models and forecast horizons. Market-input models have lower RMSE than their history-only matches in 17 of 18 comparisons, although Holm-adjusted support is concentrated in the six index pairs. Adding target history to a market-input model helps only the two terminal index specifications, and neither improvement is statistically significant. Forecast errors and revisions decline overall as resolution approaches, with local reversals, while the weighted-training counts provide a robustness check rather than another market-versus-benchmark test.

Chapter 5 uses these target-level results for the descriptive H4 and H5c comparisons and for the economic interpretation.

Chapter 5

Cross-Variable Comparison and Economic Interpretation

Chapter 4 evaluates each target separately. This chapter compares the targets for two questions: whether relative performance depends on target type (H4), and which target has the steepest decline in error during a contract's life (H5c). Neither comparison is confirmatory. The final section interprets the economic mechanisms supported by the combined evidence.

5.1 Relative Skill across the Three Targets (H4)

Raw errors are not comparable across percentage points, basis points, and index points. Each market is therefore scaled by its target-specific benchmark. Let MSE_K denote the market's mean squared error on common events and MSE_B the benchmark's error. The reported measures are

$$R = 100\sqrt{\frac{\mathrm{MSE}_K}{\mathrm{MSE}_B}}, \qquad S = 100\left(1 - \frac{\mathrm{MSE}_K}{\mathrm{MSE}_B}\right), \tag{5.1}$$

relative RMSE and the MSE skill score. These measures scale forecast errors by a reference when the series use different units (Hyndman & Koehler, 2006; Murphy, 1988). Values of $R < 100$ favour the market. Uncertainty is estimated with a paired bootstrap of R using 10,000 event-level resamples (Efron, 1979).

Table 5.1: H4: Relative Accuracy across Targets

Target	Benchmark	n	Reading	R	95% bootstrap
CPI inflation	Cleveland nowcast	9	Full ladder	79.2	[41.0, 124.4]
Federal funds rate	Fed funds futures	7	Full ladder	75.7	[49.9, 142.8]
S&P 500 close	E-mini futures	11	Top 3	137.8	[59.7, 286.4]

Notes: Kalshi accuracy relative to each target's conventional benchmark on the common window, July 2025 to May 2026. Values below 100 favour the market. Intervals are paired event bootstraps of R; each is computed against 100 for its own target, and no statistic compares one target with another.

The point estimates in Table 5.1 follow the proposed H4 ordering. On the two scheduled macroeconomic outcomes the market is ahead of the respective external benchmark,

by 42.7 percent of mean squared error for the policy rate and 37.2 percent for the CPI. On the continuously traded index the market is behind, with a skill score of −89.9 percent.

However, each 95% interval contains 100 – the value indicating equal RMSE between Kalshi and its benchmark – so none shows a statistically clear difference. The intervals also overlap substantially, leaving the cross-target ordering uncertain. With 7, 9, and 11 common events, the design has limited power.

H4 has three additional restrictions. First, the common window is shorter than each target's full test sample, so these estimates differ from those in Chapter 4. Second, the procedure produces separate intervals relative to 100 and does not test direct cross-target contrasts. Third, CPI and the policy rate use the prespecified full ladder, while the index uses the top-3 measure selected after observing the full-ladder result in Section 4.2.2. The full-ladder index result is still less favourable, so the direction is unchanged but the magnitude depends on the selected representation.

H4 is therefore descriptive and partly post-hoc. The point estimates suggest different relative performance for scheduled announcements and a continuously traded asset, but the difference is not established statistically.

5.2 Steepness of the Horizon Profile (H5c)

H5c compares the decline in error across each target's horizon grid. It is exploratory because prior evidence does not determine the ordering and contract lifetimes are not directly comparable.

Table 5.2: H5c: Horizon Decay across Targets

Target	Window	RMSE far → near	Reduction	Events far / near
Federal funds rate	1,440 h → 6 h	0.1278 → 0.0157	87.7%	12 / 12
S&P 500 close	36 h → 2 h	98.69 → 51.86	47.4%	101 / 229
CPI inflation	480 h → 6 h	0.0952 → 0.0720	24.4%	15 / 15

Notes: Fall in the full-ladder RMSE between the farthest and the nearest horizon bucket on each target's full test sample. The windows are not comparable in span; the S&P 500 event count also falls with the horizon.

The order reported in Table 5.2 is policy rate, index, and CPI. This is not a ranking of learning speed because the horizon spans differ by a factor of forty. The reduction instead measures how much uncertainty remained when each contract began. The policy-rate market starts sixty days before the decision and removes most of its error. The CPI market already contains most of its eventual information twenty days before release.

The equity comparison uses 101 events at 36 hours and 229 at two hours because not all contracts open at the longest horizon. The far-horizon value therefore refers to a subset of early-opening contracts. The 158-to-41 counts in Section 4.5.1 concern the shorter futures-common sample. The profiles are not monotonic at every adjacent cutoff, as Chapter 4 records, so the table reports the overall decline.

5.3 Economic Mechanism and Practical Implications

5.3.1 What Distinguishes the Three Targets

The combined results suggest that prediction-market value depends on the forecasting problem. In particular, it depends on the difference between the information in target history and the information dispersed among traders.

For the policy rate, information arrives on a scheduled calendar. A committee announces a specific decision at a known time, and historical rates cannot fully reveal that choice. History-only flexible models perform worse than a simple three-lag autoregression. Market error declines by a factor of eight during the contract, with most of the improvement before the final days. The market aggregates expectations about the decision rather than merely extrapolating past rates. The machine-learning models reinforce this interpretation. History-only GBM and MLP forecasts are less accurate than the AR(3), while Kalshi-input models generally outperform their matched history-only versions but remain far behind the direct market mean. Adding rate history to Kalshi inputs also worsens all six matched forecasts, suggesting that greater model flexibility adds estimation noise rather than new information in the twelve-meeting sample.

For CPI, the relevant alternative is an institutional nowcast. Historical CPI has little predictive value in this sample, so Kalshi clearly outperforms its time-series baseline. The Cleveland nowcast (Knotek & Zaman, 2017), which uses daily energy prices and other higher-frequency indicators, achieves comparable accuracy. The small sample leaves their difference unresolved. The relatively flat horizon profile is consistent with inflation inputs being observable weeks before publication. The machine-learning results point in the same direction. Every Kalshi-input model has lower RMSE than its matched history-only model, although no comparison survives multiplicity correction. Adding CPI history to Kalshi inputs worsens all six matched forecasts, and no trained model improves on the direct implied mean, indicating that the market path is informative but difficult to improve through additional modelling.

For the index, two mechanisms are separate. First, weak-form efficiency in Fama (1970) implies that the previous close should be difficult to improve. The similar no-change and AR results support this interpretation, as does the broader evidence in Welch and Goyal (2008). A small residual advantage is consistent with Grossman and Stiglitz (1980), because costly information must earn a return. Second, distant brackets may trade once and become stale (Greene, 2026). The full-ladder mean therefore mixes current and stale prices, while the top-3 measure filters toward active brackets. The difference between the readings is an aggregation problem, not evidence that the market contains no information. The machine-learning models mainly reproduce this interpretation. History-only GBM and MLP forecasts do not improve on the previous-close or AR(3) benchmarks, whereas all six Kalshi-input models significantly outperform their matched history-only versions. Twelve trained runs beat the full-ladder mean, but none beats the top-3 reading supplied through their market inputs, indicating that the models mainly reproduce the filtering of active brackets rather than discover additional predictability.

5.3.2 Accuracy without Calibration

Accurate central estimates do not imply calibrated probabilities. Point accuracy concerns the location of a distribution, while calibration concerns its shape. Conversely, Gneiting and Raftery (2007) note that calibration without sharpness can also be uninformative.

CPI is well centred but has an upper tail that is too narrow. This direction is opposite to the effect predicted by risk premia and longshot overpricing (Bürgi et al., 2025; Kitsul & Wright, 2013), suggesting errors in beliefs rather than only price formation. The index distribution is centred but too wide, consistent with those distortions. The policy-rate distribution is too concentrated for the tests to be informative. Serial independence of PIT values is not tested, so none of these findings establishes full dynamic calibration.

5.3.3 Practical Relevance

The implied mean is the most defensible direct market output, but its useful timing and appropriate benchmark differ by target. CPI forecasts lose relatively little accuracy between twenty days and six hours before release, so they can serve as an early indicator alongside the Cleveland nowcast. Policy-rate forecasts improve much more during the contract and should therefore be treated as a changing measure of expectations rather than a fixed early forecast. For the index, the most informative Kalshi reading is obtained close to the market close and from actively traded brackets; the full-ladder mean can be distorted by stale outer-bracket prices.

The benchmark results do not support treating Kalshi as a universal replacement for established forecasts. Its differences from the Cleveland nowcast and fed funds futures are unresolved in the small common samples. For the index, the full-ladder mean is less accurate than E-mini futures, while the top-3 comparison is inconclusive. Kalshi is therefore most useful as a complementary source of market expectations, with the relevant reading chosen according to the target's trading structure.

The model results also favour parsimony. No trained model improves on the direct Kalshi readings for CPI or the policy rate. For the index, trained models improve on the full-ladder mean but not on the top-3 reading supplied through their market inputs, indicating that they mainly automate the filtering of active brackets. The added computational complexity therefore provides little out-of-sample benefit beyond selecting an appropriate market representation.

Point forecasts and probabilities require separate treatment. CPI probabilities understate upper-tail risk, while index probabilities are too dispersed; the policy-rate ladder is too concentrated for the calibration tests to be strongly informative. Raw probabilities therefore require target-specific calibration before operational use, and the direction of the correction cannot be transferred from one target to another.

These conclusions concern statistical forecast accuracy rather than the profitability of a trading strategy. The analysis does not incorporate fees, bid–ask spreads, liquidity constraints, or a user's decision-specific loss. The practical contribution is consequently informational: the results identify which market reading is most credible, when it becomes informative, and when it should be combined with an external benchmark.

The main benchmark comparisons use small macroeconomic samples or shortened common windows, and several remain inconclusive. Model findings apply to the estimated architectures, training designs, and refit schedule. Chapter 6 gives the full limitations.

Chapter 6

Limitations

The limitations are organised by type rather than target because most apply to more than one analysis. Each section states the conclusion affected by the limitation.

6.1 Statistical Inference and Calibration

Macroeconomic inference rests on only 15 CPI releases, 12 policy meetings, and H4 samples of 9, 7, and 11 events. Even large RMSE differences can fail Holm-adjusted tests in short samples (Harvey et al., 1997), so failure to reject does not establish no difference. H1b also tests equal accuracy without a non-inferiority margin; its inconclusive CPI and policy-rate results do not demonstrate equivalence.

The index sample has a different problem. Its 229 test days are consecutive, and volatility clustering creates serial dependence in squared errors. The variance of mean loss differences is estimated from the sample variance only and may be understated under positive dependence. S&P 500 p-values may therefore be too small. Autocorrelation-consistent standard errors or a block bootstrap over consecutive days would address this dependence, but neither is implemented. The direction of the change is not guaranteed. Large differences are likely to remain, while the top-3 H1b comparison and two terminal H3b reversals are provisional.

H2 tests PIT uniformity and interval coverage, but not serial independence, the second condition in Diebold et al. (1998). It therefore addresses only marginal calibration, a greater restriction for 859 consecutive trading days than for events separated by weeks. Across 500 randomisation seeds, CPI and the S&P 500 reject both uniformity tests every time; the policy rate rejects the Kolmogorov–Smirnov test 19 times and the Cramér–von Mises test 26 times. Its non-rejection is common, not invariant.

6.2 External Validity and Data Coverage

All test events occur in 2025–2026, a period containing a government shutdown, two missing CPI months, faster inflation, and three anticipated rate cuts. CPI upper-tail miscalibration and the market's advantage over history-based models may therefore be period-specific.

The policy-rate test sample contains three anticipated 25-basis-point cuts but no intermeeting move, 50-basis-point test-period change, or dissent surprise. Its high modal accuracy may not generalise to emergency or contested decisions. The main labelled-

sample surprise is September 2024, when the realised 50-basis-point cut had probability 0.30. Equity evidence covers only relatively liquid same-day closing contracts on one platform, not longer contracts where staleness may matter more; platform and general market effects therefore cannot be separated.

The data contain trades rather than quotes or order-book states. Last trades can be stale, and the freshness feature records rather than removes this problem. Twelve CPI events are excluded for insufficient trading. The results describe trade-observable forecasts, not each market's full information state.

6.3 Representation and Benchmark Construction

Representation sensitivity differs sharply by target. CPI full-ladder and top-3 forecasts correlate at 0.997 and differ by 0.013 percentage points on average. For the policy rate, one poorly represented meeting makes top-3 RMSE more than four times full-ladder RMSE, although twelve events do not resolve the difference statistically. For the S&P 500, full-ladder and top-3 RMSE are 46.51 and 24.98, with $p < 0.001$. Equity H1b and H1c therefore depend on the representation.

Other choices include the monotonicity repair, tail placement half a strike spacing beyond the extremes, three-strike volume selection, the hourly grid, and assumed 10:00 Eastern Time availability of Cleveland nowcasts. H2 may depend on the tail convention, for which no alternative is evaluated.

Contract prices are interpreted as probabilities without a premium adjustment. Although theory bounds the gap in some settings (Manski, 2006; Wolfers & Zitzewitz, 2006), point forecasts and calibration statistics still mix risk-neutral and physical probabilities (Kitsul & Wright, 2013). CPI miscalibration runs opposite to the expected premium effect, whereas index miscalibration has the same direction, so tail-risk pricing and staleness cannot be separated.

The policy-rate futures procedure maps the average effective rate into target-upper-bound units using the latest pre-cutoff basis, but does not remove the futures risk premium (Piazzesi & Swanson, 2008). The residual premium may be comparable with the 0.0009 percentage point H1b difference, so the inconclusive comparison does not establish parity. The equity benchmark assumes a fixed cash–futures basis between its anchor and cutoff and applies a ninety-minute freshness rule. Licensed data cover 158 of 229 test days. The full-sample horizon profile falls from 229 events at two hours to 101 at 36 hours, while the futures-common profile falls from 158 to 41; index H1b and the two horizon profiles therefore use different samples.

6.4 Model and Design Uncertainty

Models use at most 44 CPI events and 34 policy meetings, five seeds, and fixed conservative hyperparameters. The weighted policy-rate Kalshi-input LSTM has seed RMSE of 0.244–0.392, wider than the H5b difference between training designs, and both policy-rate H5b reversals occur in recurrent models. Failure to beat the market is therefore conditional on the available sample and tested architectures.

Two choices followed inspection of the evidence. Weighted training uses half-lives of 240 hours for macroeconomic targets and 12 hours for the index without theoretical derivation or prior registration, making H5b a robustness check. The lower-error top-3

index measure was also selected after the primary full-ladder result; its ex ante filter choice is unknown.

Monthly rather than daily index refitting leaves parameters up to one month old and may disadvantage the models. On the policy rate's 25-basis-point grid, a discrete forecast can also be exactly correct while regression incurs interpolation error. Part of the model deficit may therefore be structural.

6.5 Comparability, Scope, and Further Work

The targets share hypotheses, losses, tests, and rolling origins, but not implementations. Validation blocks contain five macroeconomic events and twenty index events; refitting is per event for macroeconomic targets and monthly for the index. Horizon grids and counts differ. Cross-target conclusions are conditional on these adaptations.

H4 is descriptive and partly post-hoc for three reasons. Its common window contains only 9, 7, and 11 events; all confidence intervals cross 100; and it estimates separate target intervals rather than a direct cross-target contrast. It also uses the top-3 index measure selected after the full-ladder result, while retaining the prespecified full ladder for macroeconomic targets.

The design does not identify causal mechanisms. Matched-input comparisons locate information relative to the specified models, but show neither that the market causes accuracy nor that differences are profitable. The mechanisms in Chapter 5 are interpretations consistent with the evidence.

Further work should add proper scores that combine calibration and sharpness (Gneiting & Raftery, 2007), test PIT independence, and base equity inference on standard errors robust to autocorrelation or a block bootstrap. A prospectively specified freshness rule should be evaluated out of sample, and a trading-based study should test whether the measured inefficiencies are economically exploitable (Fama, 1970).

Chapter 7

Conclusion

This thesis evaluates the forecasting value of prices on a regulated prediction market. It studies a monthly data release, a scheduled policy decision, and a continuously traded asset price within a common framework with documented target-specific adaptations.

Kalshi outperforms the simple baseline for every target. Its inflation error is about one third of the AR(3) error, its policy-rate error is about one eighth of the no-change error, and its index error is about one fifth lower than the previous-close error. Results against stronger external benchmarks are less favourable. The Cleveland nowcast and federal funds futures comparisons are inconclusive on 15 releases and 12 meetings. Equal-accuracy tests without a non-inferiority margin cannot establish equivalence. For the index, the prespecified full-ladder RMSE is 48.58 points, compared with 20.74 for E-mini futures on the common sample. The top-3 market reading narrows this difference, but remains sensitivity evidence. Calibration is separate from point accuracy. The CPI distribution is too narrow in the upper tail, while the index distribution is too wide. Because PIT independence is not tested, these are findings about marginal calibration only.

Market-price paths contain more useful information than target history for the tested models. Market-input models outperform matched history-only models in 17 of 18 comparisons, while adding history to a market-input model improves only two. No model improves on the market price for either macroeconomic target. For the index, 12 of 18 models outperform the full ladder by filtering stale brackets, but none outperforms the top-3 reading. These findings are conditional on the tested architectures and do not identify a causal mechanism or a profitable trading strategy.

Cross-target evidence is descriptive. Relative-accuracy point estimates are better for the two scheduled outcomes and worse for the index, but all intervals cross the benchmark value, no direct cross-target statistic is estimated, and the index uses a post-hoc top-3 reading. Horizon evidence is more consistent: errors and revisions decline overall for all targets, with local reversals. The total error decline ranges from 87.7 percent for the policy rate to 24.4 percent for CPI, but this ordering is exploratory. Weighting distant windows improves all six CPI configurations and four of six configurations for each other target.

The thesis contributes a broader competitor set and a comparative design. Kalshi prices are evaluated against statistical models, external market benchmarks, and flexible models trained on the market's own history. The main limitations are the small macroeconomic samples, serial dependence in daily equity errors, and sensitivity of equity results to ladder construction. The practical conclusion is limited: the implied mean

is the most defensible direct output, while implied probabilities require target-specific calibration before use.

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Appendix A

Additional Forecast Diagnostics

This appendix supplements Figure 4.1 with terminal full-ladder error plots for the federal funds rate and the S&P 500.

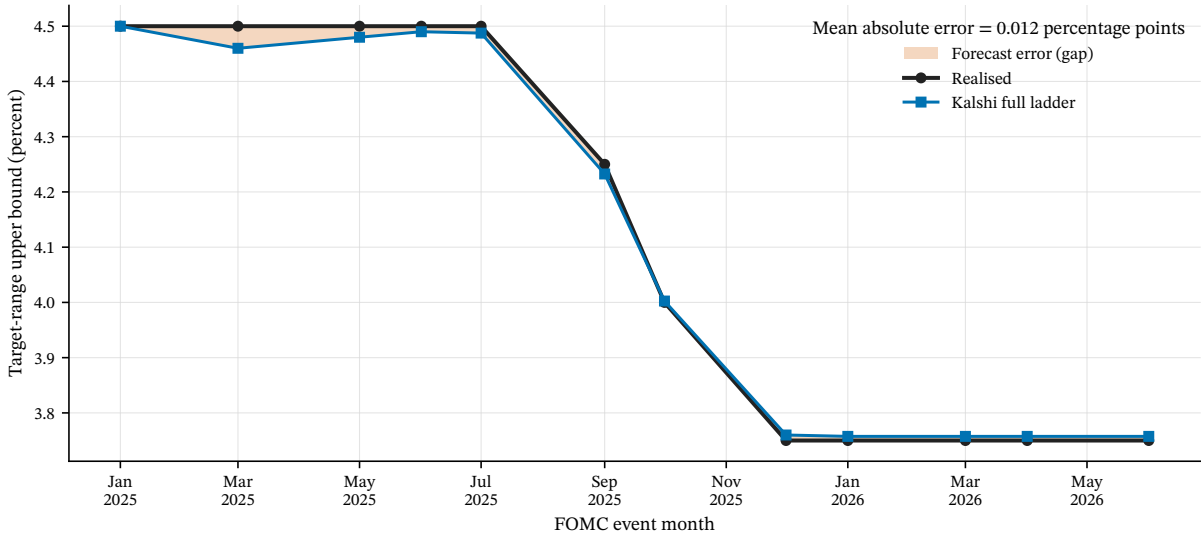


Figure A.1: Terminal full-ladder Kalshi forecast and realised federal funds target-range upper bound, 12 test meetings. Shading is the forecast error.

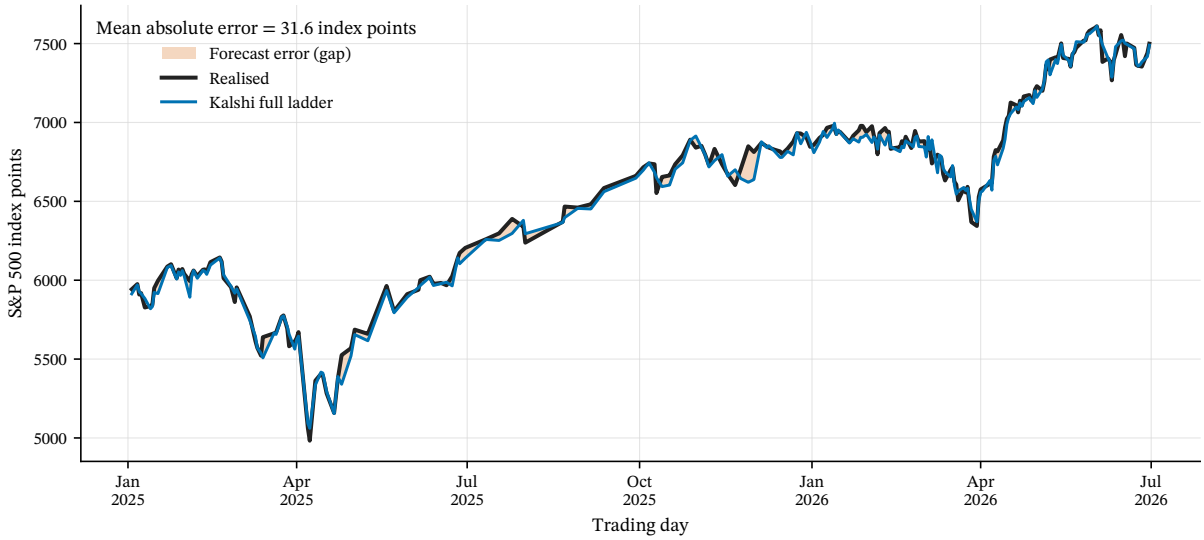


Figure A.2: Terminal full-ladder Kalshi forecast and realised S&P 500 cash close, 229 test days. Shading is the forecast error.